SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

An Autonomous Institution | Approved by AICTE | Affiliated to Anna University
Accredited by NAAC with A++ Grade, Kuniamuthur, Coimbatore - 641008

83rd Rank

DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING
(CYBER SECURITY)

AUTONOMOUS
CURRICULUM AND SYLLABI
V - VI (SEMESTERS)

REGULATIONS 2022

SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY
An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade
Kuniamuthur, Coimbatore – 641008
Phone : (0422)-2678001 (7 Lines) | Email : info@skcet.ac.in | Website : www.skcet.ac.in

Curriculum & Syllabi

Regulation 2022

2023-27 Batch

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY
An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade
Kuniamuthur, Coimbatore – 641008
Phone : (0422)-2678001 (7 Lines) | Email : info@skcet.ac.in | Website : www.skcet.ac.in

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

(Batch 2023-2027)

VISION OF THE INSTITUTION

- To Produce Globally Competitive Engineers with High Ethical Values and Social Responsibilities.

MISSION OF THE INSTITUTION

- To impart the highest quality state-of-the-art technical education by providing impetus to innovation, research, and development and empowering students with entrepreneurship skills.
- To instill ethical values, imbibe a sense of social responsibility, and strive for societal well-being.
- To identify the needs of society and offer sustainable solutions through outreach programs.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

VISION OF THE DEPARTMENT

- To be a globally renowned academic department for quality education and research in the field of Cyber security with ethical values and social commitment

MISSION OF THE DEPARTMENT

- To Impart comprehensive technical education to produce highly competent Cyber security professionals and Researchers
- Provide an academic environment with state-of-the-art technological infrastructure to provide scalable cyber security solutions
- Impart ethics, Social responsibility and necessary professional, leadership skills through student centric activities

I.	PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)
PEO 1	Get recognized as effective professionals for their applied skills, problem solving capabilities and professional skills in the field of cyber security
PEO 2	Enrich skills and adopt emerging cyber security needs to pursue life-long learning and serve the society with social concern and code of ethics

I.	PROGRAMME OUTCOMES (POs)
PO 1	**Engineering Knowledge:** Apply the knowledge of basic sciences and engineering fundamentals to solve engineering problems.
PO 2	**Problem Analysis:** Analyze the complex engineering problems and give solutions related to chemical & allied industries
PO 3	**Design/ development of solutions:** Identify the chemical engineering problems, design and formulate solutions to solve both industrial & social related problems.
PO 4	**Conduct investigations of complex problems:** Design & conduct experiments, analyze and interpret the resulting data to solve Chemical Engineering problems.
PO 5	**Modern tool usage:** Apply appropriate techniques, resources and modern engineering & IT tools for the design, modelling, simulation and analysis studies.
PO 6	**The engineer and society:** Assess societal, health, safety, legal and cultural issues and their consequent responsibilities relevant to professional engineering practice.
PO 7	**Environment and sustainability:** Understand the relationship between society, environment and work towards sustainable development.
PO 8	**Ethics:** Understand their professional and ethical responsibility and enhance their commitment towards best engineering practices.
PO 9	**Individual and team work:** Function effectively as a member or a leader in diverse teams, and be competent to carry out multidisciplinary tasks.
PO 10	**Communication:** Communicate effectively in both verbal & non-verbal and able to comprehend & write effective reports.
PO 11	**Project management and finance:** Understand the engineering and management principles to manage the multidisciplinary projects in whatsoever position they are employed.
PO 12	**Life-long learning:** Recognize the need of self-education and life-long learning process in order to keep abreast with the ongoing developments in the field of engineering

II. PROGRAMME SPECIFIC OUTCOMES (PSOs)	
The Graduates of B.E – COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY) programme will be able to:	
PSO 1	Attain the policies in information assurance and analyze the factors in an existing system and design implementations to comprehend and anticipate future challenges
PSO 2	Implement innovative cyber security solutions using standard tools, practices and technologies without compromising the privacy of individuals and entities
PSO 3	Use cyber security and cyber forensics software/tools. Design cyber-security strategies and assess cyber-security risk management policies to protect an organization's information and assets.

I.	II. MAPPING OF PEOs WITH POs											
PEO	**POs**											
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
1	3	3	3	3	2	2	–	1	2	1	1	2
2	3	3	3	3	3	2	–	–	2	–	–	2
	1- low, 2 - medium, 3 - high, ‘-' - no correlation											

III. MAPPING OF PEOs WITH PSOs			
	PSO 1	**PSO 2**	**PSO 3**
PEO 1	3	3	3
PEO 2	3	3	2

AUTONOMOUS CURRICULA AND SYLLABI
Regulations 2022(23-27 Batch)

SEMESTER I					
S. No.	**Course Code**	**Courses**	**L/T/P**	**Total Hours**	**Credits**
Theory (Internal 40 Marks & External 60 Marks)					
1	23MA101	Mathematics I	3 / 1 / 2	4	4
2	23TA101	Heritage of Tamils	1 / 0 / 0	1	1
Theory with Practicals (Internal 50 Marks & External 50 Marks)					
3	23IT101	Application Development Practices	1 / 0 / 4	5	3
4	23CS101	Problem Solving using C++	1 / 0 / 4	5	3
5	23EN101	Oral and Written Communication Skills	2 / 0 / 2	4	3
6	23CY101	Networking and Communication	3 / 0 / 2	5	4
Mandatory Course (Internal 100 Marks)					
7	23MC101	Induction Programme	Three Weeks		
			TOTAL	**24**	**18**

SEMESTER II					
S. No.	**Course Code**	**Courses**	**L/T/P**	**Total Hours**	**Credits**
Theory (Internal 40 Marks & External 60 Marks)					
1	23MA201	Mathematics II	3 / 1 / 0	4	4
2	23AS101	Applied Science	4 / 0 / 0	4	4
3	23TA201	Tamils and Technology	1 / 0 / 0	1	1
Theory with Practicals (Internal 50 Marks & External 50 Marks)					
4	23CY202	Operating Systems	3 / 0 / 2	5	4
5	23CS201	Data Structures and Algorithms	1 / 0 / 4	5	3
6	23CD201	Database Management Systems	1 / 0 / 4	5	3
7	23CY203	Programming in Java	1 / 0 / 4	5	3
8	23MC102	Environmental Sciences	1 / 0 / 0	1	0
Practicals(Internal 60 Marks & External 40 Marks)					
9	23AS101	Applied Science Laboratory	0 / 0 / 2	2	2
			TOTAL	**33**	**24**

SEMESTER III					
S. No.	**Course Code**	**Courses**	**L/T/P**	**Total Hours**	**Credits**
Theory-Blended Learning(Internal 100 Marks)					
1	23GE301	Universal Human Values	3 / 0 / 0	3	3
Theory (Internal 40 Marks & External 60 Marks)					
2	23CY301	Cyber Law and Digital Forensics	3 / 0 / 0	3	3
3	23CY302	Data Analytics and Visualization	3 / 1 / 0	4	4
4	23MA301	Mathematical Foundations for Computer Science	3 / 1 / 0	5	4
Theory with Practicals (Internal 50 Marks & External 50 Marks)					
5	23CS301	Advanced Java Programming	1 / 0 / 4	5	3
6	23IT301	Web Technology using React	1 / 0 / 4	5	3
Mandatory Course-Blended Learning (Internal 100 Marks)					
7	23MCXXX	Mandatory Course: II	2 / 0 / 0	2	0
			TOTAL	**27**	**20**

SEMESTER IV					
S. No.	**Course Code**	**Courses**	**L/T/P**	**Total Hours**	**Credits**
Theory (Internal 40 Marks & External 60 Marks)					
1	23CY403	Auditing IT Infrastructure for Compliance	3 / 1 / 0	4	4
Theory with Practicals (Internal 50 Marks & External 50 Marks)					
2	23CY401	Ethical Hacking	3 / 0 / 2	5	4
3	23CY402	Access Control and Identity management	3 / 0 / 2	5	4
4	23AD402	Design and analysis of algorithms	1 / 0 / 4	5	3
5	23IT402	Web Frameworks using REST API	0 / 0 / 4	4	2
6	23AD403	Managing Cloud and Containerization	1 / 0 / 4	5	3
7	23ME305	Design Thinking and Idea Lab	0 / 0 / 2	2	1
			TOTAL	**30**	**21**

SEMESTER V					
S. No.	**Course Code**	**Courses**	**L/T/P**	**Total Hours**	**Credits**
Theory (Internal 40 Marks & External 60 Marks)					
1	23CY501	Security Assessment and Risk Analysis	3 / 0 / 0	3	3
2	23CY502	Computational Number Theory for Cryptography	3 / 1 / 0	4	4
3	23CY503	Security Policies and Implementation	3 / 0 / 0	3	3
Theory with Practicals (Internal 50 Marks & External 50 Marks)					
4	23CY504	Cloud Security	2 / 0 / 2	4	4
5	23XXXX	Professional Elective - I	2 / 0 / 2	4	3
6	23XXXX	Professional Elective - II	2 / 0 / 2	4	3
Mini Project (Internal 100 Marks)					
7	23CY505	Application Development	0 / 0 / 6	6	3
			TOTAL	**28**	**23**

SEMESTER VI					
S. No.	**Course Code**	**Courses**	**L/T/P**	**Total Hours**	**Credits**
Theory (Internal 40 Marks & External 60 Marks)					
1	23CY601	Blockchain Technologies	3 / 0 / 0	3	3
2	23CY602	Enterprise Networking Security and Automation	3 / 0 / 0	3	3
Theory with Practicals (Internal 50 Marks & External 50 Marks)					
3	23CY603	Hacker Technique Tools and Incident Handling	3 / 0 / 2	5	4
4	23CY604	Cyber Attack Simulation and Security Testing	2 / 0 / 2	4	4
5	23XXXX	Professional Elective - III	2 / 0 / 2	4	3
6	23XXXX	Professional Elective - IV	2 / 0 / 2	4	3
Mini Project					
7	23CY605	Capstone Project	0 / 0 / 4	4	2
			TOTAL	**27**	**22**

SEMESTER VII					
S. No.	**Course Code**	**Courses**	**L/T/P**	**Total Hours**	**Credits**
Theory (Internal 40 Marks & External 60 Marks)					
1	23CYXXX	Open / Emerging/ Industrial Elective - I	3 / 0 / 0	3	3
2	23CYXXX	Open / Emerging/ Industrial Elective - II	3 / 0 / 0	3	3
3	23CYXXX	Open / Emerging/ Industrial Elective- III	3 / 0 / 0	3	3
4	23CYXXX	Professional Elective - V	1 / 0 / 4	5	3
5	23CYXXX	Professional Elective - VI	1 / 0 / 4	5	3
Project(Internal 60 Marks & External 40 Marks)					
6	23CY701	Project Phase I	0 / 0 / 6	6	3
Internship(Internal 100 Marks)					
7	23EES01	Internship	28 Days		2
			TOTAL	**25**	**20**

SEMESTER VIII					
S. No.	**Course Code**	**Courses**	**L/T/P**	**Total Hours**	**Credits**
Project(Internal 60 Marks & External 40 Marks)					
1	23CY801	Project Phase II	0 / 0 / 24	24	12
			TOTAL	**162**	

SCHEME OF CREDIT DISTRIBUTION – SUMMARY

S. No.	Stream	Credits/Semester								C	%
		I	II	III	IV	V	VI	VII	VIII		
1	Humanities & Social Sciences Including Management (HSMC)	4	7	3	1					15	9.25
2	Basic Sciences (BSC)		8							8	4.9
3	Engineering Sciences (ESC)			4						4	2.4
4	Professional Core (PCC)	10	13	13	20	14	14			84	51.85
5	Professional Electives (PEC)					6	6	6		18	11.11
6	Open/Emerging/Industry (OEC) / Emerging Elective Courses (EEC)							9		9	5.55
7.	Project Work (PROJ)					3	2	5	12	22	13.5
8.	Mandatory Course (MC) / Spoken Hindi									Non credit	
Total		14	28	20	21	23	22	20	12	160	

STRUCTURE FOR UNDERGRADUATE ENGINEERING PROGRAM			
S. No.	**Course Work - Subject Area**	**AICTE Suggested Credits**	**SKCET Credits (160)**
1.	Humanities and Social Sciences (HS), including Management;	16	15
2.	Basic Sciences (BS) including Mathematics, Physics, Chemistry, Biology;	23	8
3.	Engineering Sciences (ES), including Materials, Workshop, Drawing, Basics of Electrical/Electronics/Mechanical/Computer Engineering, Instrumentation;	29	4
4.	Professional Subjects-Core (PC), relevant to the chosen specialization/branch; (May be split into Hard (no choice) and Soft (with choice), if required	59	84
5.	Professional Subjects – Electives (PE), relevant to the chosen specialization/ branch;	12	18
6.	Open Subjects- Electives (OE), from other technical and/or emerging subject areas;	9	9
7.	Project Work, Seminar and/or Internship in Industry or elsewhere.	15	22
8.	Mandatory Courses (MC)	Non-credit	Non-credit
	Total		160
Minor Variations is allowed as per need of the respective disciplines			

HUMANITIES & SOCIAL SCIENCES INCLUDING MANAGEMENT (15 Credits)					
S. No.	**Course Code**	**Course Title**	**L/T/P**	**Contact hrs./Wk.**	**C**
1.	23TA101	Heritage of Tamils	1 / 0 / 0	1	1
2.	23EN101	Oral and Written Communication Skills	2 / 0 / 2	3	3
3.	23TA201	Tamils and Technology	1 / 0 / 0	1	1
4.	23AS101	Applied Science	4 / 0 / 0	4	4
5.	23AS102	Applied Science Laboratory	0 / 0 / 4	4	2
6.	23GE301	Universal Human Values	3 / 0 / 0	3	3
7.	23ME305	Design Thinking and Idea Lab	0 / 0 / 2	2	1

BASIC SCIENCE COURSES (8 Credits)					
S. No.	**Course Code**	**Course Title**	**L/T/P**	**Contact hrs./Wk.**	**C**
1.	23MA101	Mathematics I	3 / 1 /0	4	4
2.	23MA201	Mathematics II	3 / 1 / 0	4	4

ENGINEERING SCIENCE COURSES (04 Credits)					
S. No.	**Course Code**	**Course Title**	**L/T/P**	**Contact hrs./Wk.**	**C**
1.	23MA301	Mathematical Foundations for Computer Science	3 / 1 / 0	5	

PROFESSIONAL CORE COURSES (84 Credits)					
S. No.	**Course Code**	**Course Title**	**L/T/P**	**Contact hrs./Wk.**	**C**
1.	23IT101	Application Development Practices	1 / 0 / 4	5	3
2.	23CS101	Problem Solving using C++	1 / 0 / 4	5	3
3.	23CY101	Networking and Communication	3 / 0 / 2	4	4
4.	23CY202	Operating Systems	3 / 0 / 2	5	4
5.	23CS201	Data Structures and Algorithms	1 / 0 / 4	5	3
6.	23CD201	Database Management Systems	1 / 0 /4	5	3
7.	23CY201	Programming in Java	1 / 0 /4	5	3
8.	23CY301	Cyber Law and Digital Forensics	3 / 1 / 0	3	3
9.	23CY302	Data Analytics and Visualization	3 / 1 / 0	4	4
10.	23CS301	Advanced Java Programming	1 / 0 /4	5	3
11.	23IT301	Web Technology using React	1 / 0 /4	5	3
12.	23CY401	Ethical Hacking	3 / 0 / 2	5	4
13.	23CY402	Access Control and Identity management	3 / 0 / 2	5	4
14.	23CY403	Auditing IT Infrastructure for Compliance	3 / 1 / 0	4	4
15.	23AD402	Design and analysis of algorithms	1 / 0 / 4	5	3
16.	23IT402	Web Frameworks using REST API	0 / 0 / 4	4	2
17.	23AD403	Managing Cloud and Containerization	1 / 0 / 4	5	3
18.	23CY501	Security Assessment and Risk Analysis	3 / 0 / 0	3	3
19.	23CY502	Computational Number Theory for Cryptography	3 / 1 / 0	4	4
20.	23CY503	Security Policies and Implementation	3 / 0 / 0	3	3
21.	23CY504	Cloud Security	2 / 0 / 2	4	4
22.	23CY601	Blockchain Technologies	3 / 0 / 0	3	3
23.	23CY602	Enterprise Networking Security and Automation	3 / 0 / 0	3	3
24.	23CY603	Hacker Technique Tools and Incident Handling	3 / 0 / 2	5	4
25.	23CY604	Cyber Attack Simulation and Security Testing	3 / 0 / 2	5	4

PROFESSIONAL ELECTIVE COURSES (18 Credits)							
S. No.	**Course Code**	**Course Title**	**L/T/P**	**Contact hrs./Wk.**	**C**	**Ext/ Int**	**Cat.**
ELECTIVE STREAM I – MACHINE LEARNING ENGINEERING							
1	23AD902	Exploratory Data Analysis (EDA) using Python	2/0/2	4	3	50/50	PEC
2	23AD905	Statistical Methods and Basic Machine Learning Models	2/0/2	4	3	50/50	PEC
3	23IT901	Advanced ML Techniques	2/0/2	4	3	50/50	PEC
4	23IT902	NLP	2/0/2	4	3	50/50	PEC
5	23IT903	Computer Vision	1/0/4	5	3	50/50	PEC
6	23AD906	GenAI Advanced Prompt Engineering & LLMs	1/0/4	5	3	50/50	PEC
ELECTIVE STREAM II - DATA ANALYST WITH ML ESSENTIALS							
1	23AD901	Data Storytelling and Visualization	2/0/2	4	3	50/50	PEC
2	23AD902	Exploratory Data Analysis (EDA) using Python	2/0/2	4	3	50/50	PEC
3	23AD903	Problem solving with Analytical & Design Thinking	2/0/2	4	3	50/50	PEC
4	23AD904	PowerBI	2/0/2	4	3	50/50	PEC
5	23AD905	Statistical Methods and Machine Learning Models	2/0/2	4	3	50/50	PEC
6	23AD906	GenAI Advanced Prompt Engineering & LLMs	1/0/4	5	3	50/50	PEC
ELECTIVE STREAM III - CLOUD IT ADMINISTRATION							
1	23CS901	Implementing and Administering Enterprise Networks	2/0/2	4	3	50/50	PEC
2	23CS902	Linux System Administration	2/0/2	4	3	50/50	PEC
3	23CS903	Information Security Systems	2/0/2	4	3	50/50	PEC
4	23CS904	Low-Code No-Code Application Building	2/0/2	4	3	50/50	PEC
5	23CS905	Virtualization, Cloud Computing and SysOps	1/0/4	5	3	50/50	PEC
6	23CS906	Continuous Monitoring and Observability (AWS)	1/0/4	5	3	50/50	PEC

ELECTIVE STREAM IV – CYBER SECURITY							
1	23CS901	Implementing and Administering Enterprise Networks	2/0/2	4	3	50/50	PEC
2	23CS902	Linux System Administration	2/0/2	4	3	50/50	PEC
3	23CS903	Information Security Systems	2/0/2	4	3	50/50	PEC
4	23CY901	Cloud Computing and Containerized Virtual Infrastructure	2/0/2	4	3	50/50	PEC
5	23CY902	Penetration Testing	1/0/4	5	3	50/50	PEC
6	23CY903	Security Operations of Information systems	1/0/4	5	3	50/50	PEC
ELECTIVE STREAM V – FULL STACK SOFTWARE DEVELOPMENT (for Minor)							
1	23CS911	Managing and Querying Database (RDBMS) MySQL / Postgre SQL	2/0/2	4	3	50/50	PEC
2	23CS912	Java / Python: Object-Oriented Programming	2/0/2	4	3	50/50	PEC
3	23IT911	Web Development and UI/UX Essentials	2/0/2	4	3	50/50	PEC
4	23IT912	Build Single-Page Applications using React	2/0/2	4	3	50/50	PEC
5	23IT913	Build Back-end Application using Spring Boot / FAST API	1/0/4	5	3	50/50	PEC
6	23AD906	GenAI Advanced Prompt Engineering & LLMs	1/0/4	5	3	50/50	PEC

INDIAN KNOWLEDGE SYSTEM (02 Credits)					
SL. No.	**Course Code**	**Course Title**	**L/T/P**	**Contact hrs./Wk.**	**C**
1.	23EES01	Internship	28 Days		2

OPEN/ EMERGING/ INDUSTRY (09 Credits)					
S. No.	**Course Code**	**Course Title**	**L/T/P**	**Contact hrs./Wk.**	**C**
1.	23CY001	Cyber Security Fundamentals	3 / 0 / 0	3	3
2.	23CY002	Cryptography – Tools and Techniques	3 / 0 / 0	3	3
3.	23CY003	Cyber security Auditing and Assurance	3 / 0 / 0	3	3
4.	23CY004	Web Security	3 / 0 / 0	3	3
5.	23CY005	Operating System Security	3 / 0 / 0	3	3
6.	23CY006	Security Governance, Risk, and compliance	3 / 0 / 0	3	3

PROJECT WORK (22 Credits)					
S. No.	**Course Code**	**Course Title**	**L/T/P**	**Contact hrs./Wk.**	**C**
1.	23CY505	Application Development	0 / 0 / 6	6	3
2.	23CY605	Capstone Project	0 / 0 / 4	4	2
3.	23CY701	Project Phase - I	0 / 0 / 6	6	3
4.	23CY801	Project Phase - II	0 / 0 / 24	24	12

PROFESSIONAL ELECTIVE COURSES: VERTICALS				
Vertical I - ML Engineering	**Vertical II - Data Analyst with ML Essentials**	**Vertical III - Cloud IT Administration**	**Vertical IV - Cybersecurity**	**Vertical V – Full Stack Software Development (Minor)**
Exploratory Data Analysis (EDA) using Python	Data Storytelling and Visualization	Implementing and Administering Enterprise Networks	Implementing and Administering Enterprise Networks	Managing and Querying Database (RDBMS) MySQL / Postgre SQL
Statistical Methods and Basic Machine Learning Models	Exploratory Data Analysis (EDA) using Python	Linux System Administration	Linux System Administration	Java / Python: Object-Oriented Programming
Advanced ML Techniques	Problem solving with Analytical & Design Thinking	Information Security Systems	Information Security Systems	Web Development and UI/UX Essentials
NLP	PowerBI	Low-Code No-Code Application Building	Cloud Computing and Containerized Virtual Infrastructure	Build Single-Page Applications using React
Computer Vision	Statistical Methods and Machine Learning Models	Virtualization, Cloud Computing and SysOps	Penetration Testing	Build Back-end Application using Spring Boot / FAST API
GenAI Advanced Prompt Engineering & LLMs	GenAI Advanced Prompt Engineering & LLMs	Continuous Monitoring and Observability (AWS)	Security Operations of Information systems	GenAI Advanced Prompt Engineering & LLMs

INTERN (02 Credits)					
S. No.	**Course Code**	**Course Title**	**L/T/P**	**Contact hrs./Wk.**	**C**
1.	23EES01	Internship	28 Days		2

VALUE ADDED COURSES (Based on student's interest)							
S. No	**Course Code**	**Course Title**	**L/T/P**	**Contact hrs./Wk.**	**C**	**SDG Mapping**	**Sem**
1.	23VA101	PEGA Certification	1 / 0 / 0	1	1	4, 9,12	III
2.	23VA102	Amazon Web Service (AWS)	1 / 0 / 0	1	1	4, 9,12	IV
3.	23VA103	Google Certified Cloud Professional (GCCP)	1 / 0 / 0	1	1	4, 9,12	V
4.	23VA104	CISCO Certification	1 / 0 / 0	1	1	4, 9,12	VI
5.	23VA105	Wipro PRP Certification	1 / 0 / 0	1	1	4, 9,12	VI

MANDATORY COURSES (Non-Credits) (Courses conducted either by internal faculty or through MOOCs)					
S. No.	**Course Code**	**Course Title**	**L/T/P**	**Contact hrs./Wk.**	**C**
1.	23MC101	Induction Programme	0		
2.	23MC102	Environmental Sciences	1 / 0 / 0	1	0

L: Lecture **T:** Tutorial **P:** Practical **C:** Credit **O:** Outside Class hours **Cat.:** Category

HSMC : Humanities and Social Sciences including Management
BSC : Basic Science Courses
ESC : Engineering Science Courses
PCC : Professional Core Courses
PEC : Professional Elective Courses
OEC : Open and Emerging Elective Courses
PRJ : Project Work
INT : Internship
MC : Mandatory Course

Definition of Credit:

L – Lecture	1 Hr. Lecture (L) per week	1 credit
T – Tutorial	1 Hr. Tutorial (T) per week	1 credit
P - Practical/Practice (Project and Industry based Courses)	1 Hr. Practical (P) per week	0.5 credit

SEMESTER – V

23CY501	SECURITY ASSESSMENT AND RISK ANALYSIS	3/0/0/3
Nature of Course:	C (Theory Concept)	
Pre requisites:	Nil	

Course Objectives:

1.	Understand secure software design principles and architecture.
2.	Apply risk management strategies to mitigate software security risks.
3.	Learn cryptographic methods and authentication models in enterprise systems.
4.	Analyze security challenges in enterprise systems, including e-commerce platforms.
5.	Develop skills for security testing and quality assurance

Course Outcomes

Upon completion of the course, students shall have ability to

C501.1	Understand the principles of secure software design and its architecture.	[U]
C501.2	Apply risk management frameworks to address software security risks.	[AP]
C501.3	Implement cryptographic techniques and authentication models.	[AP]
C501.4	Analyze security architectures and identify vulnerabilities.	[A]
C501.5	Apply security testing to analyze and ensure software reliability.	[AP]

Course Contents:

MODULE I SECURE SOFTWARE DESIGN AND ARCHITECTURE 15

Computer security, principles of secure software, trusted computing base, etc, threat modeling, advanced techniques for mapping security requirements into design specifications. Secure software implementation, deployment and ongoing management. Software design and an introduction to hierarchical design representations. Difference between high-level and detailed design. Handling security with high-level design. General Design Notions. Security concerns design at multiple levels of abstraction, Design patterns, quality assurance activities and strategies that support early vulnerability detection, Trust models, security Architecture & design reviews.

MODULE II SOFTWARE ASSURANCE AND RISK MANAGEMENT 15

Software Assurance Model: Identify project security risks & selecting risk management strategies, Risk Management Framework, Security Best practices/ Known Security Flaws, Architectural risk analysis, Security Testing & Reliability (Penn testing, Risk Based Security Testing.

MODULE III ENTERPRISE SOFTWARE SECURITY AND CRYPTOGRAPHY 15

Software Security in Enterprise Business: Identification and authentication, Enterprise Information Security, Symmetric and asymmetric cryptography, including public key cryptography, data encryption standard (DES), advanced encryption standard (AES), algorithms for hashes and message digests. Authentication, authentication schemes, access control models, Kerberos protocol, public key infrastructure (PKI), protocols specially designed for e-commerce and web applications, firewalls and VPNs. Security development frameworks. Security issues associated with the development and deployment of information systems, including Internet-based e-commerce, e-business, and e-service systems.

TOTAL PERIODS 45

Text Books:	
1.	W. Stallings, Cryptography and network security: Principles and practice, 5 th Edition, Upper Saddle River, NJ: Prentice Hall., 2011
2.	C. Kaufman, r. Perlman, & M. Speciner, Network security: Private communication in a public world, 2 nd Edition, Upper Saddle River, NJ:Prentice HalL, 2002
3.	C. P. Pfleeger, S. L. Pfleeger, Security in Computing, 4 th Edition, Upper Saddle River, NJ:Prentice Hall, 2007.
Reference Books:	
1.	T4. M. Merkow, & J. Breithaupt, Information security: Principles and practices. Upper Saddle River, NJ:Prentice Hall, 2005.
2.	Gary McGraw, Software Security: Building Security In, Addison-Wesley, 2006.
Web References:	
1.	https://csrc.nist.gov
2.	https://owasp.org
3.	https://www.microsoft.com/en-us/securityengineering/sdl
4.	https://www.coursera.org/learn/software-security

23CY502	COMPUTATIONAL NUMBER THEORY FOR CRYPTOGRAPHY	3/1/0/4
Nature of Course:	C (Theory Concept)	
Pre requisites:	Nil	

Course Objectives:

1.	To understand and learn a complete introduction to some basic concepts and results in abstract algebra and elementary number theory.
2.	To identify the algorithms for primality testing, with an emphasis on the Miller-Rabin test, the elliptic curve test, and the AKS test.
3.	To Learn and apply algorithms such as trial division, Pollard's ρ method, elliptic curve method, and quadratic sieve for factorization.
4.	To Develop the ability to solve discrete logarithm problems using advanced techniques.
5.	To understand and analyze the principles of secret-key cryptography, focusing on RSA, and elliptic curve-based schemes.

Course Outcomes

Upon completion of the course, students shall have ability to

C502.1	Understand and learn a complete introduction to some basic concepts and results in abstract algebra and elementary number theory.	[R]
C502.2	Identify the algorithms for primality testing, with an emphasis on the Miller-Rabin test, the elliptic curve test, and the AKS test.	[A]
C502.3	Learn and apply algorithms such as trial division, Pollard's ρ method, elliptic curve method, and quadratic sieve for factorization.	[AP]
C502.4	Solve discrete logarithm problems using advanced techniques.	[A]
C502.5	Understand and analyze the principles of secret-key cryptography, focusing on RSA, and elliptic curve-based schemes.	[A]

Course Contents:

MODULE I ELEMENTARY OPERATIONS 20

Fundamentals-BasicAlgebraicStructures-DivisibilityTheory-ArithmeticFunctions - Congruence Theory - Primitive Roots -Elliptic Curves. Complexity of basic operations like additions, multiplications for integers and polynomials. Computational Number Theory: Basic Tests - Miller–Rabin Test- Elliptic Curve Tests - AKS Test.

MODULE II INTEGER FACTORIZATION AND DISCRETE LOGARITHMS 20

Basic Concepts- Trial Divisions Factoring -ρ and $p - 1$ Methods - Elliptic Curve Method - Continued Fraction Method - Quadratic Sieve. Discrete Logarithms: Basic Concepts- Baby-Step Giant-Step Method – Pohling – Hellman Method – Index Calculus – Theory of Elliptic Curves - Elliptic Curve Discrete Logarithms.

MODULE III ELLIPTIC CURVE MASSEY-OMURA CRYPTOGRAPHY

Secret-Key Cryptography: Cryptography and Cryptanalysis-Classic Secret-Key C Cryptography - Modern Secret-Key Cryptography. Integer Factorization Based Cryptography - RSA Cryptography - Elliptic Curve Discrete Logarithm Based Cryptography: Basic Ideas - Elliptic Curve Diffie–Hellman–Merkle Key Exchange Scheme - Elliptic Curve Massey.

TOTAL PERIODS: 60

Text Books:	
1.	Song Yan, "Computational Number Theory and Modern Cryptography", Wiley Publications (P), Higher Education Press, UK,2013.
2.	Victor Shoup A*, "Computational Introduction to Number Theory and Algebra",* Cambridge University Press,2008.
3.	Joachim Von Zur Gathen, Jurgen Gerhard, "Modern Computer Algebra", Cambridge University Press,2008
Reference Books:	
1.	Bach & Shallit, "Algorithmic Number Theory", The MIT Press
2.	Rudolf Lidl,Harald, "Encyclopedia of Mathematics and Its Applications Finite Fields", Cambridge University Press, 2003
Web References:	
1.	https://nptel.ac.in/courses/106103015/
2.	https://people.csail.mit.edu/madhu/FT05/
3.	https://onlinecourses.nptel.ac.in/noc25_cs14/preview
4.	https://www.coursera.org/learn/number-theory-cryptography

23CY503	SECURITY POLICIES AND IMPLEMENTATION		3/0/0/3
Nature of Course:	C (Theory Concept)		
Pre requisites:	Nil		
Course Objectives:			
1.	To gain knowledge of Security Policy Frameworks		
2.	To identify potential security threats and vulnerabilities and security problems in current software issues		
3.	To describe security policies to address evolving challenges and technological advancements.		
4.	To examine the importance of documentation for policy implementation and enforcement.		
5.	To be able to implement the best practices of software security Policies..		
Course Outcomes			
Upon completion of the course, students shall have ability to			
C503.1	Identify the role of an information systems security (ISS) policy framework in overcoming business challenges.	[AP]	
C503.2	Understand the relationship between regulatory compliance requirements and information system security policies.	[U]	
C503.3	Explain how security policies help mitigate risks and support business processes in various domains of a typical IT infrastructure.	[U]	
C503.4	Describe how to design, organize, implement, and maintain IT security policies.	[AP]	
C503.5	Identify the issues related to security policy implementations and the keys to success.	[AP]	

Course Contents:

MODULE I NEED OF IT SECURITY POLICY FRAMEWORKS 15

Introduction to Information Systems Security, Information Assurance, Governance, Information Systems Security Policies, Creating Policies, The Seven Domains of IT Infrastructure, Security Policies mitigate within the Seven Domains. Information Security Policy Implementation Issues: The role of Human Resources Policies**,** Policy Roles, Responsibilities and accountability.

MODULE II TYPES OF POLICIES AND APPROPRIATE FRAMEWORKS 15

Introduction, Program Framework Policy, Information Assurance Considerations, Information Systems Security Considerations. User Domain Policies: The weakest Link in the information Security Chain, Seven Types of Users, Acceptable Use policy (AUP), The Privileged Level Access Agreement (PAA), Security Awareness Policy (SAP). IT Infrastructure Security Policies: Workstation Domain Policy, LAN Domain Policies, LAN –to-WAN Domain Policies, WAN Domain Policies, Remote Access Domain Policies, System/Application Domain Policies, Telecommunication Domain Policies, Cloud Security Domain Policies. Data Classification and Handling Policies and Risk Management Policies: Data Classification Policies, Data Handling Policies, Risk Assessment Policies. Incident Response Team (IRT) Policies: Introduction, Business Impact Analysis (BIA) Policies, Procedures for Incident Response, Disaster Recovery Plan Policies.

MODULE III IMPLEMENTING AND MAINTAINING AN IT SECURITY POLICY FRAMEWORK	**15**

IT Security Policy Implementations – Target state, Executive Buy-in, Cost and Impact, policy language, Employee Awareness and Training, Information Dissemination, Policy Implementation Issues, Governance and Monitoring. IT Security Policy Enforcement: Organizational Support for IT Security Policy Enforcement, An Organizational Right to Monitor User Actions and Traffic, Legal Implications of IT Security Policy Enforcement. IT Policy Compliance and Compliance Technologies Creating a Baseline Definition for Information Systems Security, Automated IT Security Policy Compliance, Compliance Technologies and Solutions

TOTAL PERIODS: 45

Text Books:

1.	Robert Johnson, Chuck Easttom, “Security Policies and Implementation Issues”, 3rd Edition, Jones & Bartlett Learning,2022.
2.	Thomas R. Peltier, “Information Security Policies and Procedures”, 2nd Edition, Auerbach,2003.

Reference Books:

1.	Sari Greene, “Security Policies and Procedures: Principles and Practices”, 1st Edition, Pearson Publishing, 2005.
2.	William Stallings, “Network Security Essentials", Pearson Education, 6th Edition, 2018.

Web References:

1.	https://www.nist.gov/cyberframework
2.	https://www.iso.org/isoiec-27001-information-security.html
3.	https://www.sans.org/reading-room
4.	https://csrc.nist.gov/publications/sp
5.	https://www.securitymagazine.com/

23CY504	CLOUD SECURITY	2/0/2/4
Nature of Course:	(Theory, Analytical)	
Prerequisites:	Network security and cryptography and cloud computing	

Course Objectives:

1.	Understand the importance of security in Cloud Computing.
2.	Understand the cloud security and privacy issues
3.	Understand the Threat Model and Cloud Attacks
4.	Understand the Data Security and Storage
5.	Analyze Security Management in the Cloud.

Course Outcomes:

Upon completion of the course, students shall have ability to:

C504.1	Appreciate the importance of security in a cloud environment	[U]
C504.2	Distinguish the various cloud security and privacy issues.	[U]
C504.3	Analyze the various threats and Attack tools	[A]
C504.4	Able to understand the Data Security and Storage	[U]
C504.5	Analyze the Security Management in the Cloud.	[A]

Course Contents:

Module I IMPORTANCE OF CLOUD SECURITY 10

Security Concepts in Cloud, Confidentiality, privacy, integrity, authentication, Defense in depth, least privilege, Importance of Security in PaaS, IaaS , SaaS, TaaS User authentication in the cloud, Cryptographic Systems, Symmetric cryptography, stream ciphers, block ciphers, modes of operation Multi-Tenancy issues, X.509 certificates, OpenSSL

Module II CLOUD SECURITY AND PRIVACY ISSUES 10

Introduction, Cloud Security Goals/Concepts, Cloud Security Issues, Security Requirements for Privacy, Privacy Issues in Cloud. Security over FoG and edge computing. Infrastructure Security: The Network Level, the Host Level, the Application Level, SaaS, PaaS, IaaS, TaaS Application Security,

Module III THREAT MODEL AND CLOUD ATTACKS 10

Introduction, Threat Model- Type of attack entities, Attack surfaces with attack scenarios, A Taxonomy of Attacks, Attack Tools-Network-level attack tools, VMlevel attack tools, VMM attack tools, Security Tools, VMM security tools. Security Management in the Cloud- Security Management Standards, Availability Management, Access Control, Security Vulnerability, Patch, and Configuration Management.

TOTAL PERIODS(Theory) 30

LIST OF EXPERIMENTS

1. Install Virtualbox/VMware Workstation with different flavours of Linux or Windows OS on top of Windows 7 or 8.
2. Install a C compiler in the virtual machine created using VirtualBox and execute simple programs.
3. Install Google App Engine. Create *hello world* app and other simple web applications using Python/Java.
4. Use GAE launcher to launch the web applications.
5. Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim.
6. Find a procedure to transfer the files from one virtual machine to another virtual machine.
7. Find a procedure to launch a virtual machine using TryStack (Online OpenStack Demo Version).
8. Install Hadoop single node cluster and run simple applications like wordcount.

TOTAL PERIODS(Lab) 30

TOTAL PERIODS 60

Text Books:

1	Cloud Security Attacks, Techniques, Tools, and Challenges by Preeti Mishra, Emmanuel S Pilli, Jaipur R C Joshi Graphic Era, 1st Edition published 2022 by CRC press.
2	Cloud Computing with Security Concepts and Practices Second Edition by Naresh Kumar Sehgal Pramod Chandra, P. Bhatt John M. Acken, 2 nd Edition Springer nature Switzerland AG 2020
3	Cloud Security and Privacy by Tim Mather, Subra Kumaraswamy, and Shahed Lati First Edition, September 2019.

Reference Books:

1	Essentials of Cloud Computing by K. Chandrasekaran Special Indian Edition CRC press.
2	Cloud Computing Principles and Paradigms by Rajkumar Buyya, John Wiley.

Web References:

1	https://www.hcltechsw.com/bigfix/secure-infrastructur automation? utm_source= google&utm_medium=cpc&utm_campaign
2	https://www.cisa.gov/sites/default/files/publications/Cloud%20Security%20Technical%20Reference%20Architecture.pdf

23CY505	APPLICATION DEVELOPMENT	0/0/6/3
Nature of Course	F (Theory Programming)	
Pre-Requisite	Cloud Computing	

Course Objectives:

1	To discuss the essence of front-end development skills.
2	To impart the knowledge of React components used in Spring boot development platforms.
3	Ability to understand and use Setup Cloud API.
4	To deploy and test the React App used in Spring Boot.
5	To learn the Spring Cloud concepts using Docker.

Course Outcomes: Upon completion of the course, students shall have ability to:

C505.1	Identify the basic concepts and design issues of React.	[R]
C505.2	Understand the principles of process and Spring boot.	[U]
C505.3	Illustrate the approaches in scheduling and Spring Cloud to apply in real world problems.	[AP]
C505.4	Apply concepts of Micro services Communication to the issues that occur in Real time applications.	[AP]
C505.5	Identify issues related to Docker, API Gateway.	[AP]
C507.1	Examine common React, Availability and Scalability.	[A]

Course Contents:

REACT INTRODUCTION **30**

Components, Routes, State, Props, hooks, Higher Order Functions, Axios and Services, Ant Design. Redux: Core Concept, Data Flow, Store, Actions, Pure function, Reducers, Devtools, Middleware, Webpack, Redux Integration. Spring boot: Annotations, Beans, Configuration, HTTP Methods, Crud, Postman Overview. Spring Security: Authentication, Authorization, Security Implementation. Configure Security, Authentication Manager, HTTP Security, Circular Reference Error. JWT Implementation: JWT Overview, JWT Libraries, Helper Methods, Token Generation and Validation, Implementing JWT Authorization, Filter. OAUTH Implementation: Introduction, Sample flow, Authorization code grant type flow, Implicit grant flow, Password Grant Type flow, Client, Credential Grand type flow, Refresh token Grand type flow, Validating token,Oauth2 integration with Spring Security. Building Micro services : Monolith Architecture and Challenges of Monolith Architecture,What is Micro services & How It Solves the Challenges of Monolith Architecture, Micro services Architecture Benefits and Best Practices, Understanding Spring Cloud and It's Important Modules,Micro service Applications and It's Port Mapping

MICROSERVISES COMMUNICATION OVERVIEW **30**

Micro services Communication using Rest Template, Micro services Communication using Web Client, Micro services Communication using Spring Cloud Open Feign - Understanding service Registry – Spring Cloud Netflix Eureka Server Implementation, Update on Using Spring Boot 3 Version, Register Micro service as Eureka Client, Update on using Spring Boot 3 Version, Register Micro service as Eureka Client, Running Multiple Instances of Micro service, Load Balancing with Eureka, Open Feign and Spring Cloud Load Balancer API gateway using Spring Cloud gateway: Understanding API Gateway - Create and Set up API Gateway Micro service, Update on Using Spring Boot 3 Version, Register API-Gateway as Eureka Client to Eureka Server, Configuring API Gateway Routes and Test using Postman Client, Using Spring Cloud Gateway to Automatically Create Rout.

CENTRALIZED CONFIGURATIONS USING SPRING CLOUD CONFIG SERVER **30**

How to Use Spring Cloud Config Server, Create and Setup Spring Cloud Config Server Project in IntelliJ IDEA, Update on Using Spring Boot 3 Version, Register Config-Server as Eureka Client, Set up Git Location for Config Server, Refactor Department-Service to use Config Server, Refactor Employee-Service to use Config Server, Refresh Use case - No Restart Required After Config Changes,REACT Frontend Micro service: Create React App using Create React App Tool, Adding Bootstrap in React Using NPM, Write HTTP Client Code to Connect React App with API-Gateway (REST API Call), Create a React Component and Integrate with API Gateway Microservice, RabbitMQ Core Concepts: RabbitMQ Architecture, Install and Setup RabbitMQ using Docker, Explore RabbitMQ using RabbitMQ Management UI, Create and Setup Spring Boot 3 Project in IntelliJ, Connection Between Spring Boot and RabbitMQ,, Configure RabbitMQ in Spring Boot Application, Create RabbitMQ Producer, Create REST API to Send Message, Create RabbitMQ Consumer,, Configure RabbitMQ for JSON Message Communication, Create RabbitMQ Producer to Produce JSON Message, Create REST API to Send JSON Object, Create RabbitMQ Consumer to Consume JSON Message,, Dockering Spriing boot App : Install Docker Desktop, General Docker Workflow, Create Spring Boot Project and Build Simple REST API, Create Docker file to Build Docker Image, Build Docker Image from Dockerfile,

Course Guidelines:

1. Students choose a project topic from a list of approved options or propose their own idea from the area specified in the content and Faculty Coordinator/guide approval required for student-proposed projects
2. Number of students in the project team should be maximum of 4 and Every student shall have a project guide.
3. The entire semester shall be utilized by the students to do their project work by receiving the directions from the project guide.
4. Teams should submit a project proposal, including objectives, scope, timeline, and resources and Faculty Coordinator/guide reviews and approves the proposal.
5. Students should choose projects in line with the Departmental Mission, Vision, and Program Outcomes.

6. Students should attend periodic reviews to present the progress of the project to faculty and peers' team and Evaluation is based on project outcomes, presentation quality, and teamwork.
7. Teams submit a final project report, including results, conclusions, and recommendations as specified in the guidelines issued by the COE.
8. Students should not be involved in unethical behavior, such as plagiarism, copyright violations, etc while working on projects and when submitting project reports.
9. Every student team will be required to prepare and submit two (2) copies plus (no. of students) copies of the Project report of typical length 30 – 60 pages (excluding Appendices).

A final external project viva-voce examination will be conducted to evaluate the student's Individual and team performance based on project outcomes, presentations, reports, and teamwork by an Internal and External Examiners.

TOTAL PERIODS	**90**

Text Books:

1	Merih Taze, "Engineers Survival Guide: Advice, tactics, and tricks after a decade of working at Facebook, Snapchat", Microsoft Paperback – November 28, 2021.
2	Gerardus Blokdyk, "Secure Microservices A Complete Guide", Edition Paperback – July 17, 2021.
3	Theo H King, "Aws: The Ultimate Guide from Beginners to Advanced For the Amazon Web Services", (2020 Edition), Paperback – Import, 21 December 2019.

Reference Books:

1	Craig zacker, "Exam ref pl-900 Microsoft power platform", paperback – 8 February 2021

Web References:

1	https://awscloud.in/

SEMESTER – VI

23CY601	**BLOCK CHAIN TECHNOLOGIES**	**3/0/0/3**
Nature of Course:	C (Theory Concept)	
Pre requisites:	Ethical Hacking	

Course Objectives:

1	To Understand the foundational principles of blockchain technology
2	To apply the principles and features of Distributed Ledger Technology (DLT)
3	To Evaluate the structure and functionality of smart contracts
4	To Differentiate between various blockchain ecosystems
5	To Explore advanced blockchain protocols and token economies

Course Outcomes

Upon completion of the course, students shall have ability to

C601.1	Explain the architecture, design principles, and challenges of blockchain technology	[U]
C601.2	Illustrate the structure and functioning of Distributed Ledger Technologies.	[AP]
C601.3	Demonstrate the development and lifecycle of smart contracts, and apply them in domain-specific use cases like healthcare and property transfer using DLT platforms	[UP]
C601.4	Analyze different types of blockchain ecosystems and their governance models by identifying roles, interactions, and strategic implications of ecosystem components	[A]
C601.5	Differentiate and analyze blockchain protocols, token types, and token economy structures, with emphasis on legal frameworks and their application in blockchain-based projects.	[A]

Course Contents:

Foundations of Blockchain **15**

Blockchain Architecture – Challenges – Applications – Blockchain Design Principles -The Blockchain Ecosystem - The consensus problem - Asynchronous Byzantine Agreement - AAP protocol and its analysis - peer-to-peer network – Abstract Models - GARAY model - RLA Model - Proof of Work (PoW) - Proof of Stake (PoS) based Chains - Hybrid models.Origin of Ledgers – Types and Features of Distributed Ledger Technology (DLT) - Role of Consensus Mechanism - DLT Ecosystem - Distributed Ledger Implementations

Smart Contracts **15**

Blockchain - Ethereum - Public and Private Ledgers – Registries – Ledgers - Practitioner Perspective: Keyless Technologies, Transparency as a Strategic Risk, Transparency as a StrategicAsset, Usage of Multiple IDs - Zero Knowledge Proofs - Implementation of Public and Private Blockchain- Anatomy of a Smart Contracts - Life Cycle - Usage Patterns - DLT-based smart contracts -Use Cases: Healthcare Industry and Property Transfer - Decentralization versus Distribution - Centralized-distributed (Ce-Di) organizations -Decentralized-distributed (De-Di) organizations - Decentralized Autonomous Organizations: Aragon, DAOstack, DAOhaus and Colony.

Blockchain Ecosystem and Protocols	**15**
One-Leader Ecosystem - Joint Venture or Consortia Ecosystems - Regulatory Blockchain Ecosystems - Components in Blockchain Ecosystem: Leaders, Core Group, Active Participants, Users, Third-Party Service Providers - Governance for Blockchain Ecosystems- Ethereum tokens – Augur - Golem - Understanding Ethereum tokens - App Coins and Protocol Tokens - Blockchain Token Securities Law Framework - Token Economy - Token sale structure - Ethereum Subreddit	
TOTAL PERIODS 45	

Text Books:

1.	Dhillon, V., Metcalf, D., and Hooper, M, Blockchain enabled applications, 2017, 1st Edition, CA: Apress, Berkeley.
2.	Imran Bashir, “Mastering Blockchain - A technical reference guide to the inner workings of blockchain, from cryptography to DeFi and NFTs”, Packt Publishing Ltd, Fourth Edition, 2023

Reference Books:

1.	Diedrich, H., Ethereum: Blockchains, digital assets, smart contracts, decentralized autonomous organizations, 2016, 1st Edition, Wildfire publishing, Sydney.
2.	2. Wattenhofer, R. P, Distributed Ledger Technology: The Science of the Blockchain (Inverted Forest Publishing), 2017, 2nd Edition, Createspace Independent Pub, Scotts Valley, California, US

Web References:

1.	https://www.ibm.com/topics/blockchain
2.	https://ethereum.org/en/developers/docs/smart-contracts/
3.	https://www.sciencedirect.com/science/article/pii/S0160791X21001747
4.	https://consensys.net/knowledge-base/defi/what-are-daos/
5	https://cointelegraph.com/learn/what-is-proof-of-stake-in-cryptocurrency

23CY602	**ENTERPRISE NETWORKING SECURITY AND AUTOMATION**	**3/0/0/3**
Nature of Course:	C (Theory Concept)	
Pre requisites:	Ethical Hacking	

Course Objectives:

1	To configure advanced routing and switching protocols.
2	To resolve common issues with routing and switching protocols and identify threats to enhance network security.
3	To implement IPv4 Access Control Lists (ACLs).
4	To configure Network Address Translation (NAT) services.
5	To explain virtualization, software-defined networking, and automation

Course Outcomes

Upon completion of the course, students shall have ability to

C602.1	Explain and apply the configuration procedures for advanced routing and switching protocols.	[AP]
C602.2	Analyse and resolve routing and switching issues while identifying security threats.	[A]
C602.3	Demonstrate the implementation of IPv4 ACLs for network security.	[AP]
C602.4	Configure and verify NAT services to support IPv4 address scalability.	[AP]
C602.5	Explain the concepts of virtualization, software-defined networking, and automation.	[U]

Course Contents:

MODULE I: SINGLE-AREA OSPFv2 & NETWORK SECURITY **15**

OSPF Concepts – Features, Link-State Operation, Packet Types, Neighbor Adjacencies, Database Synchronization, DR/BDR Election, Router ID, Configuration, Network Types, and Verification. OSPF Optimization – Failover, Recovery, and Default Route Propagation. Cybersecurity Basics – Network Attacks, Threat Actors, Reconnaissance, Access Attacks, DoS/DDoS, IP Vulnerabilities, ICMP Attacks, Address Spoofing, TCP/UDP Vulnerabilities. Cryptography & ACLs – Data Confidentiality, Hashing, Encryption, Packet Filtering, ACL Types.

MODULE II: NAT, WAN, VPN, QoS, AND NETWORK MANAGEMENT **15**

NAT for IPv4 – Characteristics, Address Space, Terminology, Types, NAT64, Advantages, Configuration, and Verification. WAN Concepts – Devices, Connectivity Options, and Internet-Based Solutions. VPN and IPsec Concepts – VPN Technology, Benefits, Types, IPsec Protocols, Encryption, Authentication, and Secure Key Exchange. QoS Concepts – Network Performance, Traffic Prioritization, Network Management – Device Discovery, Time Synchronization, SNMP, Syslog, Router & Switch Maintenance, and IOS Image Management.

MODULE III: NETWORK DESIGN, TROUBLESHOOTING, VIRTUALIZATION & AUTOMATION	**15**
Network Design – Scalable networks, hierarchical models, redundancy, failure domains. Network Troubleshooting – Documentation, methodologies, tools, and layered diagnostic approaches. Network Virtualization – Cloud computing, virtualization, controllers , Network Automation – Automation fundamentals, data formats, APIs, configuration management, Cisco DNA Centre. Case Study – Implementing automation in a virtualized enterprise, analysing scalability, security, and efficiency. .	
	TOTAL PERIODS : 45

Text Books:

1.	Enterprise Networking, Security, and Automation Companion Guide (CCNAv7), by Cisco Networking Academy, Cisco Press, July 2020.
2.	CCNA 200-301 Portable Command Guide, 5th Edition, by Cisco Networking Academy, Cisco Press, 2020.

Reference Books:

1.	Routing TCP/IP, Volume 1 – by Jeff Doyle and Jennifer Carroll
2.	Network Warrior – by Gary A. Donahue (O'Reilly Media)
3.	IPv6 Essentials – by Silvia Hagen (O'Reilly Media)

Web References:

1.	https://community.cisco.com/t5/routing/need-advise-for-ospf-failover-design/td-p/5154500
2.	https://www.cisco.com/c/en/us/support/docs/ip/network-address-translation-nat/217208-understanding-nat64-and-its-configuratio.html
3.	https://www.cloudflare.com/learning/network-layer/what-is-ipsec/
4	https://irkr.fei.tuke.sk/AplikaciePocitacovychSieti/Prednasky/ENSA_Module_9%20-%20QoS%20Concepts.pdf
5	https://www.ciscopress.com/articles/article.asp?p=2189637&seqNum=4

23CY603	**HACKER TECHNIQUES TOOLS AND INCIDENT HANDLING**	**3/0/2/4**
Nature of Course:	C (Theory Concept)	
Pre requisites:	Ethical Hacking	

Course Objectives:

1	To understand Hacker Techniques and Incident Handling
2	To secure Wireless and Web Networks Against Hacker Tactics
3	To understand the Defensive Strategies for Linux Penetration Testing and Social Engineering

Course Outcomes

Upon completion of the course, students shall have ability to

C603.1	Interpret the role and significance of Public Key Infrastructure (PKI) in ensuring secure communication. (Module 1)	[U]
C603.2	Evaluate the risks associated with open ports and services on a network and prioritize them based on their potential impact. (Module 1)	[A]
C603.3	Explain the vulnerabilities associated with wireless LANs, including eavesdropping, spoofing, and man-in-the-middle attacks. (Module 2)	[U]
C603.4	Utilize techniques such as SQL injection and cross-site scripting (XSS) to demonstrate vulnerabilities in web servers and web applications. (Module 2)	[AP]
C603.5	Utilize incident response plans and procedures to effectively respond to cybersecurity incidents, including identifying and containing security breaches, preserving evidence, and restoring systems and data. (Module 3)	[AP]

Course Contents:

Module – I EXPLORING HACKER TECHNIQUES **15**

Hacker Techniques Tools and Incident Handling - Ethical Hacking and Penetration Testing – Exploring the OSI Reference Model – Symmetric Encryption – Asymmetric Encryption – Purpose of Public Key Infrastructure – Hashing – Physical Security and its Controls – Port Scanning – Identifying access machines – Mapping open ports – Enumeration and Computer System Hacking

Module – II WIRELESS & WEB SECURITY: HACKER TACTICS **15**

The importance of wireless security – Securing Bluetooth – Securing Wireless LANs – Threats to Wireless LANs – Wireless Hacking Tools – Protecting Wireless networks – Attacking Web Servers – Examining a SQL Injection – Vandalizing Web Servers – Database Vulnerabilities – Malware – Sniffers, Session Hijacking and Denial of Service Attacks

Module – III DEFENSIVE STRATEGIES: NAVIGATING LINUX PENETRATION TESTING AND SOCIAL ENGINEERING **15**

Linux Penetration Testing – Social Engineering – Types of Social Engineering attacks – Technology and Social Engineering – Best practices for Passwords – Incident Response and Defensive Technologies – Incident Response Plans – Planning for Disaster Recovery – Evidence Handling and Administration – Intrusion Detection Systems – The purpose of firewalls – Honeypots – The role of Controls – Security Best Practices.

TOTAL PERIODS(Theory) 45

LAB EXERCISES

Implement the following]

1. Generate key pairs, encrypt a file using public key, and decrypt using the corresponding private key.
2. Utilize hashing libraries such as MD5, SHA-256.
3. Scan a network to identify active hosts and open ports, analyze the results, and understand potential vulnerabilities.
4. Use enumeration tools like enum4linux, NSE scripts in Nmap.
5. Use Wi-Fi access points and tools like Aircrack-ng.
6. Configure and secure a wireless network using WPA2, detect and mitigate rogue access points.
7. Perform attacks like SQL injection, cross-site scripting (XSS), and understand the impact of vulnerable web applications.
8. Analyze malware behavior, identify persistence mechanisms, and understand how malware communicates with command and control servers.
9. Perform penetration testing on Linux systems, exploit vulnerabilities, and harden the system against attacks.
10. Simulate social engineering attacks, analyze human vulnerabilities, and develop countermeasures such as security awareness training.

TOTAL PERIODS(Lab) 30
TOTAL PERIODS 75

Text Books:

1.	Sean Philip Oriano, Michael G. Solomon, “Hacket Techniques, Tools, and Incident Handling”, Jones and Bartlett Learning, 2020,
2.	Dafydd Stuttard, Marcus Pinto, "The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws”, 2011.

Reference Books:

1.	Michael Sikorski, Andrew Honig, “Practical Malware Analysis: The Hands-On Guide to Dissecting Malicious Software", 2012.
2.	David Kennedy, Jim O'Gorman, Devon Kearns, “Metasploit: The Penetration Tester's Guide", 2011.
3.	N. K. McCarthy, “The Incident Response Pocket Guide", 2010.

Web References:

1.	https://intellicomp.net/it-services-blog/hacking-methods/
2.	https://www.rapid7.com/fundamentals/types-of-attacks/
3.	https://www.computerworld.com/article/2563639/wireless-hacking-techniques.html
4	https://www.udemy.com/topic/ethical-hacking/
5	https://www.sans.org/cyber-security-courses/hacker-techniques-incident-handling/

23CY604	**CYBER ATTACK SIMULATION AND SECURITY TESTING**	**3/0/2/4**
Nature of Course:	C (Theory Concept)	
Pre requisites:	Ethical Hacking	

Course Objectives:

1	To understand the foundational concepts of ethical hacking and the legal, ethical, and regulatory frameworks guiding penetration testing practices (e.g., ISO 27001, NIST, HIPAA).
2	To analyze and apply industry-standard security assessment methodologies such as PTES, OSSTMM, and NIST SP 800-115 for conducting effective penetration tests.
3	To perform advanced reconnaissance and exploitation techniques on networks, web applications, wireless systems, and cloud infrastructures using tools like Nmap, Metasploit, and SET.
4	To investigate post-exploitation techniques including privilege escalation, credential dumping, and persistence strategies to simulate real-world attacker behavior.
5	To develop professional penetration test reports aligned with CVSS scoring, MITRE ATT&CK framework, and organizational compliance requirements for both technical and executive stakeholders.

Course Outcomes

Upon completion of the course, students shall have ability to

C604.1	Analyze and perform reconnaissance, enumeration, and fingerprinting techniques for offensive security	[U]
C604.2	Exploit system and web-based vulnerabilities using professional-grade tools and frameworks	[A]
C604.3	Simulate red team operations targeting modern network, cloud, and wireless infrastructures	[U]
C604.4	Generate detailed technical and management-level penetration test reports with proper scoring and mitigation	[AP]
C604.5	Apply cybersecurity laws, ethics, and compliance standards to penetration testing practices in real scenarios	[AP]

Course Contents:

MODULE I FUNDAMENTALS OF PENETRATION TESTING AND TARGET ENUMERATION **15**

Introduction to cyber offense and ethical hacking- Legal, ethical, and regulatory aspects (ISO 27001, NIST 800-53, PCI-DSS, HIPAA)- Types of penetration testing: white-box, black-box, and grey-box approaches-Security assessment methodologies: PTES (Penetration Testing Execution Standard), OSSTMM, NIST SP 800-115. Open Source Intelligence (OSINT) techniques: Shodan, Maltego, Google Dorking-Active reconnaissance: Nmap scanning, Masscan, Netcat usage, and banner grabbing-Service enumeration: SMB, SNMP, FTP, SMTP, and HTTP fingerprinting-Threat intelligence collection and digital footprint mapping

MODULE II VULNERABILITY EXPLOITATION TECHNIQUES AND WIRELESS ATTACKS **15**

Exploiting network services and protocols (SMB, RDP, SSH misconfigurations)-Web exploitation techniques: SQL Injection, Cross-Site Scripting (XSS), Remote Code Execution (RCE), Cross-Site Request Forgery (CSRF)-Password-based attacks: Brute-force, dictionary attacks, rainbow table usage-Post-exploitation steps: privilege escalation, credential dumping, persistence mechanisms- WEP/WPA cracking, Evil Twin attacks, deauthentication attacks- Social engineering: phishing, spear-phishing, payload creation and delivery using SET (Social Engineering Toolkit)-Cloud penetration testing: IAM role manipulation, insecure storage exploitation (AWS S3, Azure Blob), and credential abuse-API hacking and mobile penetration testing introduction

MODULE III REPORTING AND ADVANCED RED TEAM OPERATIONS **15**

Vulnerability scoring and classification: CVSS (v3.1), risk heatmaps, OWASP risk rating-Red teaming lifecycle: planning, execution, reporting, and evasion techniques-Adversary emulation using MITRE ATT&CK framework and the Cyber Kill Chain model-Report writing for executive and technical audiences: compliance and business alignment

TOTAL PERIODS(Theroy) 45

LAB EXERCISES:

Implement the following]

1. Setting Up a Penetration Testing Lab Environment

Objective: Configure a controlled test environment for safe exploitation.

- Tools: VirtualBox/VMware, Kali Linux, Metasploit, OWASP Juice Shop, DVWA
- Tasks: Install Kali Linux and vulnerable targets, verify network communication, snapshot environment
- Outcome: A sandboxed, repeatable environment for ethical testing

2. Passive and Active Reconnaissance

Objective: Perform reconnaissance using OSINT and scanning techniques.

- Tools: Shodan, Maltego, Nmap, Netcat, Whois, nslookup
- Tasks: Map target IPs, enumerate open ports and services, analyze metadata
- Outcome: Create a target profile using digital footprint analysis

3. Vulnerability Assessment and Exploitation of Web Applications

Objective: Discover and exploit vulnerabilities in web apps.

- Tools: Burp Suite, OWASP ZAP, Nikto, SQLMap
- Tasks: Perform SQL Injection, XSS, and CSRF attacks on DVWA or Juice Shop
- Outcome: Understand OWASP Top 10 vulnerabilities and practical exploitation

4. Password Cracking and Privilege Escalation

Objective: Conduct brute-force attacks and escalate privileges.

- Tools: Hydra, John the Ripper, rockyou.txt, Linux Exploit Suggester
- Tasks: Crack SSH login, enumerate users, exploit SUID/GTFObins
- Outcome: Gain root access from a low-privilege shell

5. Wireless Network Penetration Testing

Objective: Audit and exploit Wi-Fi security protocols.

- Tools: Aircrack-ng, Wifite, Reaver
- Tasks: Capture handshake, crack WPA password, set up Evil Twin access point
- Outcome: Demonstrate security flaws in wireless encryption and authentication

6. Social Engineering and Phishing Simulation

Objective: Simulate real-world phishing attacks.

- Tools: Social Engineering Toolkit (SET), Gophish
- Tasks: Create fake login pages, clone legitimate websites, craft spear-phishing emails
- Outcome: Measure success rate of phishing campaigns and user awareness

7. Cloud Penetration Testing Basics

Objective: Identify and exploit misconfigurations in cloud environments.

- Tools: AWS/Azure free-tier sandbox, ScoutSuite, Pacu, S3Scanner
- Tasks: Enumerate IAM policies, test S3 bucket access, privilege escalation
- Outcome: Understand cloud-specific security risks and red team opportunities

8. Full-Scope Red Team Simulation and Reporting

Objective: Execute end-to-end penetration test with formal reporting.

- Tools: All previously used tools + report templates
- Tasks: Recon, exploit, maintain access, exfiltrate data, document findings
- Outcome: Submit a professional pentest report with CVSS ratings and remediation steps

TOTALPERIODS(Lab) 30
TOTALPERIODS 75

Text Books:	
1.	Penetration Testing Demystified: A Hands-on Introduction and Guide by J. Smith, A. Ray, and M. Chen, CyberSec Lab Series. 1st Edition, published 2024 by Amazon Digital Publishing.
2.	Learn Penetration Testing by Rishalin Pillay and Philip Ingram, 1st Edition, published 2024 by Packt Publishing.
Reference Books:	
1.	Penetration Testing: A Complete Guide – 2024 Edition by Gerardus Blokdyk, CyberEdge Press. 1st Edition, published 2024 by 5STARCooks.
2.	The Hacker Playbook 3: Practical Guide to Penetration Testing by Peter Kim, Secure Planet. 3rd Edition, published 2024 by Secure Planet, LLC.
Web References:	
1.	https://it.wikipedia.org/wiki/Offensive_Security?utm
2.	https://en.wikipedia.org/wiki/OWASP?utm
3.	https://pentest-tools.com/
4	https://academy.tcm-sec.com/p/learn-penetration-testing-free?utm

23AD902	EXPLORATORY DATA ANALYSIS USING PYTHON		2/0/2/3
Nature of Course	F (Theory Programming)		
Prerequisites	-		
Course Objectives:			
1	To utilize NumPy and Pandas for data manipulation and structuring.		
2	To implement querying and transforming data using Data Frame operations.		
3	To familiarize learners with descriptive and exploratory visualizations using Python libraries.		
4	To teach statistical concepts like central tendency, dispersion, and correlation in the context of EDA.		
5	To enable learners to apply best practices of univariate, bivariate, and multivariate analysis using Python and GenAI tools.		
Course Outcomes: Upon completion of the course, students shall have ability to:			
C902.1	Apply NumPy and Pandas to organize and manipulate structured data for analysis.	[U]	
C902.2	Apply Query and transform Data Frames using filters, aggregations, and CRUD operations.	[AP]	
C902.3	Create descriptive and exploratory visualizations using Matplotlib and Python libraries.	[AP]	
C902.4	Apply descriptive statistics and explore central tendency, dispersion, and correlation.	[AP]	
C902.5	Develop a comprehensive EDA project that demonstrates multivariate analysis, integrates SQL data sources, and uses GenAI tools for insights.	[AP]	

DATA STRUCTURING WITH PYTHON LIBRARIES **10**

Array Structures - 1D/2D NumPy Arrays, Indexing, Slicing, Broadcasting. Data Structures - Series, DataFrames, Data Types, Null Handling. Data Manipulation - Sort, Drop, Apply, Map, Merge, Join, groupby, concat. Data Types and Conversion – astype(), to_numeric(). Basic Data Cleaning – dropna(), fillna(), replace().

QUERY TRANSFORMATION DATA AND DESCRIPTIVE ANALYSIS **10**

Data Filtering - Conditional Filters, Logical Operators, Query Method. Aggregation - Mean, Sum, Count, Min, Max, Custom Aggregations. CRUD Operations - Insert, Update, Replace, Delete rows/columns. SQL Integration – Sqlite3, Connectors, Executing Queries from Python.

Descriptive Analysis - Mean, Median, Mode, Range, Standard Deviation, Variance. Correlation Analysis - Pearson Coefficient, Covariance Matrix. Visualization: Matplotlib Charts, Seaborn Charts, Histograms, Boxplots, Scatterplots, Pairplots.

EDA BEST PRACTICES AND MULTIVARIATE ANALYSIS **10**

EDA Techniques - Univariate, Bivariate, Multivariate Analysis. GenAI Tools - AI-generated Summary, Code Assistance, Automated Charts. Project and Summative Assessment - Perform a complete exploratory data analysis using NumPy, Pandas, and Matplotlib. Integrate SQL datasets and use GenAI tools for generating summaries, visuals, and insights.

List of Experiments

Create and manipulate 1D and 2D NumPy arrays for basic numeric operations.
Construct Pandas Series and Data Frames from lists, dictionaries, and CSV files.
Apply filtering and aggregation techniques on Data Frames using groupby and pivot table.
Perform insert, update, and delete operations on Data Frames to simulate CRUD logic.
Connect Python with a SQLite database and retrieve data into a Data Frame.
Calculate mean, median, mode, standard deviation, and correlation on sample datasets.
Create visualizations like bar charts, histograms, scatter plots, and box plots using Matplotlib.
Perform univariate and bivariate analysis on a dataset using appropriate visuals and stats.
Demonstrate multivariate analysis using pair plots or heatmaps on complex datasets.
10. Complete a mini project that uses GenAI tools to generate exploratory insights and visual summaries.

Total Theory	**30 periods**
Total Lab	**30 periods**
Total	**60 periods**

Text Books:	
1	“Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython” by Wes McKinney, Third Edition, O'Reilly Media, 2022.
2	“Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python” by Peter Bruce, Andrew Bruce, and Peter Gedeck, Second Edition, O'Reilly Media, 2020.
Reference Books:	
1	“Data Science from Scratch: First Principles with Python” by Joel Grus, Second Edition, O'Reilly Media, 2019.
2	“Python Data Science Handbook: Essential Tools for Working with Data” by Jake VanderPlas, O'Reilly Media, 2016.
3	“Hands-On Exploratory Data Analysis with Python: Perform EDA Techniques to Understand, Summarize and Investigate your Data” by Suresh Kumar Mukhiya and Usman Qamar, Packt Publishing, 2020.

23AD905	STATISTICAL METHODS AND BASIC MACHINE LEARNING MODELS	2/0/2/3
Nature of Course	F (Theory Programming)	
Prerequisites	-	

Course Objectives:

1	To develop a foundational understanding of probability theory and sampling methods essential for statistical inference.
2	To apply and interpret hypothesis testing techniques to validate assumptions using real-world datasets.
3	To perform data preparation and preprocessing for effective use in machine learning applications.
4	To explore and analyze regression and classification models for identifying patterns and making predictions.
5	To gain practical experience in implementing clustering algorithms to discover hidden structures in data.

Course Outcomes:

Upon completion of the course, students shall have ability to:

C905.1	Understand and apply probability concepts and sampling techniques for statistical inference.	[AN]
C905.2	Analyse data distributions and perform hypothesis testing using Z-test, T-test, and Chi-Square test	[AN]
C905.3	Prepare and transform data effectively for building machine learning models.	[AP]
C905.4	Apply linear regression techniques to analyse relationships between variables	[C]
C905.5	Implement classification models like logistic regression and K-Nearest Neighbors for predictive analytics and perform clustering using K-Means through a course-end project simulating real-world practices.	[AP]

PROBABILITY, SAMPLING, AND STATISTICAL INFERENCE **10**

Probability and Prediction- Basic probability- Joint and conditional probability- Real-world probability applications-Sampling and Normal Distributions- Sampling techniques- Sampling distribution- Properties of the normal distribution-Central Limit Theorem- Statistical Inference and Hypothesis Testing(Parametric) - Formulating hypotheses, Z-test and T-test for means, p-value interpretation, confidence intervals. Hypothesis Testing(Nonparametric) - Chi-Square goodness-of-fit test, test for independence in contingency tables.

DATA PREPARATION AND SUPERVISED LEARNING MODELS **10**

Data Preparation for Machine Learning- Handling missing values- Encoding categorical variables- Data normalization and scaling-Train-test split-Regression Models-Simple linear regression: line of best fit, correlation vs causation, residual analysis -Multiple linear regression: multicollinearity, variable selection, coefficient interpretation, model assumptions-Classification Models- Logistic regression: binary classification, sigmoid function, threshold tuning- Confusion matrix, ROC-AUC.

UNSUPERVISED LEARNING AND MODEL EVALUATION **10**

K-Nearest Neighbor (KNN) - Distance metrics, choosing k, model training and prediction, performance evaluation. K-Means Clustering - Centroid initialization, distance calculation, elbow method, cluster interpretation and visualization. Model Evaluation -Accuracy, precision, recall, F1-score, cross-validation- Project / Summative Assessment-Implementation of regression, classification, and clustering models- Use of real-world datasets- Model analysis and reporting.

LIST OF LAB EXPERIMENTS

Probability Estimation: Simulate and compute simple, joint, and conditional probabilities using Python.
Sampling and Visualization: Apply simple random and stratified sampling on datasets and visualize distributions.
Normal Distribution Analysis: Plot and interpret normal distribution with Python and identify z-scores.
Hypothesis Testing - Z and T Test: Perform one-sample and two-sample z-test and t-test on numerical data.
Chi-Square Test: Conduct Chi-square goodness of fit and independence tests for categorical variables.
Data Cleaning and Scaling: Perform handling of missing values, outlier detection, and standardization.
Simple Linear Regression: Implement and evaluate a linear regression model using one independent variable.
Multiple Linear Regression: Build a model using multiple features and interpret coefficients and R^2.
Classification using Logistic Regression and KNN: Train and evaluate logistic regression and KNN models using classification metrics.
K-Means Clustering Project: Apply clustering on a real dataset, use elbow method to select k, and interpret results.

Total Theory	**30 periods**
Total Lab	**30 periods**
Total	**60 periods**

Text Books:	
1	“Statistics for Data Science and Business Analysis” by Bhushan Patil, BPB Publications, 2020.
2	“Introduction to Machine Learning with Python: A Guide for Data Scientists” by Andreas C. Müller and Sarah Guido, O'Reilly Media, 2016.
Reference Books:	
1	“Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python” by Peter Bruce, Andrew Bruce, and Peter Gedeck, Second Edition, O'Reilly Media, 2020.
2	“Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow” by Aurélien Géron, Third Edition, O'Reilly Media, 2022.
3	“Python Machine Learning” by Sebastian Raschka and Vahid Mirjalili, Third Edition, Packt Publishing, 2019.

23IT901	ADVANCED MACHINE LEARNING TECHNIQUES		2/0/2/3
Nature of Course	D (Theory Application)		
Pre requisites	-		
Course Objectives:			
1.	To equip learners with the knowledge of advanced supervised learning techniques for accurate classification tasks.		
2.	To develop the ability to select optimal models using cross-validation and robust performance metrics.		
3.	To introduce ensemble learning methods for improved generalization and predictive performance.		
4.	To provide hands-on experience in unsupervised learning using hierarchical and density-based clustering methods.		
5.	To enable learners to perform dimensionality reduction and explore smart tools like GenAI for automating ML workflows.		
Course Outcomes			
Upon completion of the course, students shall have ability to			
C901.1	Illustrate advanced supervised learning techniques like Naive Bayes, SVM, and Decision Trees for data classification tasks.		[U]
C901.2	Apply unsupervised learning techniques using Hierarchical Clustering and DBSCAN to discover patterns in unlabelled data.		[AP]
C901.3	Identify ensemble learning techniques such as Random Forest, Boosting, and Stacking for improved predictive accuracy.		[AP]
C901.4	Examine dimensionality reduction techniques like PCA and LDA to enhance model performance and interpretability.		[AN]
C901.5	Develop automation of machine learning workflows using Generative AI and smart ML tools for streamlined development.		[AP]

SUPERVISED AND UNSUPERVISED LEARNING TECHNIQUES **10**

Naive Bayes Classification - Probabilistic modelling, prior and likelihood, Gaussian and Multinomial Naive Bayes, real-world applications. Support Vector Machines (SVM) - Hyperplane and margins, kernel tricks, soft vs hard margin, model tuning. Decision Trees for Classification - Tree construction, entropy, Gini index, overfitting and pruning.
Hierarchical Clustering - Dendrograms, agglomerative vs divisive, linkage criteria, cluster interpretation. DBSCAN Clustering - Density-based clustering, core points, epsilon-neighbourhood, handling noise/outliers.

MODEL EVALUATION AND ENSEMBLE LEARNING **10**
Model Validation and Selection - K-Fold, hold-out, stratified K-Fold, LOO cross-validations, model performance comparison. Random Forest & Ensemble Learning - Bagging, feature importance, OOB error, Random Forest classifier. Boosting and Stacking Models - AdaBoost basics, model stacking architecture.

DIMENSIONALITY REDUCTION TECHNIQUES, ML WORKFLOW AUTOMATION AND GENAI TOOLS **10**
Dimensionality Reduction Techniques - PCA for variance preservation, LDA for class separation. ML Workflow Automation - GenAI-powered model recommendation tools, Auto ML pipelines, smart feature engineering.

Total Periods **30**

Laboratory Component:

List of Experiments

1. Naive Bayes Classifier: Apply Gaussian/Multinomial Naive Bayes for text or numeric classification.
2. SVM Implementation: Train and test an SVM model with different kernels (linear, RBF).
3. Decision Tree Classifier: Build a tree-based classifier and visualize decision rules.
4. Hierarchical Clustering: Use dendrograms to visualize clusters and interpret structure.
5. DBSCAN Clustering: Implement density-based clustering and tune epsilon and minPts.
6. Cross-Validation: Perform k-fold cross-validation to compare multiple models.
7. Random Forest Model: Train and evaluate a random forest classifier on a structured dataset.
8. Boosting Techniques: Apply AdaBoost or XGBoost to improve model performance.
9. PCA and LDA: Apply PCA and LDA to reduce dimensions and plot transformed data.
10. AutoML Tools/GenAI Integration: Use GenAI toolkits (e.g., Google Vertex AI, AutoML, or GenAI notebooks) for automated model selection and optimization.

Total Hours: 30

Text Books:	
1.	Christopher M. Bishop, “Pattern Recognition and Machine Learning”, Springer, 2006
2.	Aurélien Géron, 2. “Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow”, Third Edition, O'Reilly Media, 2022
Reference Books:	
1.	Kevin P. Murphy, “Machine Learning: A Probabilistic Perspective”, MIT Press, 2012
2.	Andreas C. Müller, Sarah Guido, “Introduction to Machine Learning with Python: A Guide for Data Scientists”, O'Reilly Media, 2016
3.	Sebastian Raschka, Vahid Mirjalili, “Python Machine Learning”, Third Edition, Packt Publishing, 2019

23IT902	NATURAL LANGUAGE PROCESSING	2/0/2/3
Nature of Course	D (Theory Application)	
Pre requisites	-	

Course Objectives:

1.	To introduce learners to statistical methods for analysing and interpreting time series data.
2.	To equip students to build and evaluate time series forecasting models using classical methods like AR, MA, ARIMA, SARIMA, and SARIMAX.
3.	To enable learners to integrate Generative AI tools to enhance forecasting accuracy and automate predictions.
4.	To provide knowledge of Natural Language Processing (NLP) techniques to clean, process, and analyze unstructured text data.
5.	To train learners to implement text vectorization, rule-based matching, and various text classification models including fastText.

Course Outcomes

Upon completion of the course, students shall have ability to

C902.1	Interpret statistical characteristics of time series data for effective modeling and forecasting.	[U]
C902.2	Build classical time series models such as AR, MA, ARMA, ARIMA, SARIMA, and SARIMAX.	[AP]
C902.3	Apply preprocessing and cleaning of unstructured textual data using NLP techniques	[AP]
C902.4	Examine rule-based matching, vectorization, and classification for text-based applications.	[AN]
C902.5	Analyse the impact of text classification models including binary, multi-class, and multi-label using fastText and other tools.	[AN]

TIME SERIES FUNDAMENTALS & CLASSICAL FORECASTING MODELS **10**

Time Series Statistics and Exploration - Time-dependent features, trend, seasonality, autocorrelation, stationarity, ACF and PACF. AR and MA Models - Autoregressive and Moving Average models, lag selection, parameter tuning. ARMA and ARIMA Models - Combining AR and MA, differencing for stationarity, model diagnostics, error metrics. Seasonal Models (SARIMA & SARIMAX) - Seasonal components, exogenous variables, parameter selection, forecasting.

UNDERSTAND NLP FUNDAMENTALS AND TEXT PREPROCESSING **10**

Text Processing Fundamentals - Tokenization, stop word removal, stemming, lemmatization, part-of-speech tagging. Text Cleaning and Annotation -Removing noise, entity recognition, labelling strategies. Rule-Based NLP - Pattern matching with spaCy, custom rules, phrase matching, entity ruler.

TEXT CLASSIFICATION MODELS AND USE OF GENAI FOR NLP **10**

Text Vectorization Techniques - Bag-of-Words, TF-IDF, word embeddings, feature selection for modeling. Text Classification – Basics: Binary, multi-class, and multi-label approaches, confusion matrix, evaluation metrics. Advanced Classification with fastText - Text preprocessing, training classifiers, model tuning. GenAI for NLP and Forecasting - Apply GenAI to automate text preprocessing and enhance the extraction of insights from unstructured data.

Total Periods(Theory) | **30**

Laboratory Component:

List of Experiments

1. Time Series Analysis: Plot and interpret components like trend, seasonality, and residuals.
2. Stationarity Testing: Apply ADF test and visualize rolling statistics.
3. AR/MA Modeling: Build and evaluate Autoregressive and Moving Average models.
4. ARIMA/SARIMA Modeling: Apply ARIMA and SARIMA models with proper order selection.
5. SARIMAX Implementation: Incorporate exogenous variables into SARIMA models.
6. GenAI for Forecasting: Use Generative AI tools to automate and improve time series predictions.
7. Text Cleaning and Tokenization: Preprocess raw textual data with annotation and normalization.
8. Rule-Based Matching: Implement phrase and pattern matching using rule-based NLP techniques.
9. Vectorization Techniques: Convert text to numerical format using CountVectorizer or TF-IDF.
10. Text Classification: Build and evaluate binary, multi-class, and multi-label classifiers using fastText, use GenAI for NLP.

Total Periods (Lab): 30
Total Periods : 60

Text Books:	
1.	Galit Shmueli, Kenneth C. Lichtendahl Jr, “Practical Time Series Forecasting with R: A Hands-On Guide”, 2^{nd} Edition, 2016
2.	Daniel Jurafsky, James H. Martin, “Speech and Language Processing”, International Edition, 2008.
References:	
1.	Wes McKinney, “Python for Data Analysis”, 2^{nd} Edition, O'Reilly Media, 2017
2.	Steven Bird, Ewan Klein, Edward Loper, “Natural Language Processing with Python”, O'Reilly Media, 2009.
3.	Online GenAI documentation (OpenAI, Google Vertex AI) for workflow automation

23IT903	COMPUTER VISION		1/0/4/3
Nature of Course	F (Theory Programming)		
Pre requisites			
Course Objectives:			
1.	To introduce learners to the foundational principles of deep learning and its application to real-world predictive problems.		
2.	To build proficiency in developing deep learning architectures using TensorFlow or PyTorch.		
3.	To enable learners to apply LSTM and BERT models for textual analysis such as recommendation systems and sentiment classification.		
4.	To teach core computer vision techniques including image processing, classification, detection, and segmentation using CNNs.		
5.	To encourage the use of Generative AI tools for critique, interpretation, and explainability in deep learning models.		
Course Outcomes			
Upon completion of the course, students shall have ability to			
C903.1	Outline core concepts and architectures of deep learning for predictive modeling using TensorFlow or PyTorch.		[U]
C903.2	Apply LSTM and BERT models for building recommendation engines and performing sentiment analysis on text data.		[AP]
C903.3	Develop computer vision techniques including image processing, classification, and segmentation.		[AP]
C903.4	Utilize Convolutional Neural Networks (CNNs) to classify images and detect patterns in visual data.		[AP]
C903.5	Make use of Generative AI tools to improve model critique, interpretation, and explainability in computer vision tasks.		[AP]

DEEP LEARNING FUNDAMENTALS AND TEXT MODELS **5**

Deep Learning Foundations - Neurons, activation functions, forward and backward propagation. Core Building Blocks: Dense layers, dropout, batch normalization, loss functions, optimizers. Predictive Modeling with DL Model definition, training deep networks, evaluation techniques.
Text-Based DL Models - Sentiment analysis using BERT, tokenizer, transformer architecture, fine-tuning. Sequential Models – LSTM - Recurrent layers, building a recommendation system, sequence prediction.

IMAGE PROCESSING & CNNs **5**

Image Processing Fundamentals - OpenCV operations, grayscale conversion, edge detection, image transformation. Image Classification with CNN - Filters, feature maps, pooling layers, CNN architecture, model training.

TRANSFER LEARNING & DETECTION **5**

Transfer Learning for Images - Pretrained models, feature extraction, fine-tuning. Object Detection Techniques, Bounding boxes, real-time detection basics. Semantic Segmentation - Pixel-wise classification, U-Net architecture, real-time applications. Generative AI Integration - Utilize Generative AI to streamline neural network design and hyperparameter tuning, accelerating the development of deep learning models. DL Tools and Frameworks - TensorFlow/PyTorch, Keras usage; debugging, model saving.

Total Periods (Theory)	**15**

Laboratory Component:

List of Experiments

1. Build a basic neural network using TensorFlow or PyTorch to predict numeric output.
2. Perform sentiment classification using pre-trained BERT on a Twitter dataset.
3. Build a recommendation engine using LSTM with sequential user-item data.
4. Load and preprocess images using OpenCV and apply edge detection filters.
5. Construct and train a CNN from scratch on a dataset like CIFAR-10 or Fashion-MNIST.
6. Implement transfer learning using a pretrained model like ResNet or VGG16 for image classification.
7. Detect objects in an image using bounding boxes and a YOLO or SSD approach.
8. Perform semantic segmentation using U-Net and visualize segmentation maps.
9. Use Grad-CAM or similar techniques to interpret CNN predictions visually.

10. Build a GenAI-powered notebook that critiques model performance and explains misclassifications. 11. Compare image classification accuracy using CNNs vs Transfer Learning techniques. 12. Present a course-end Project using a full pipeline from problem statement to deployment-ready model.	
Total Periods (Lab) 60 **Total Periods :75**	

Text Books:	
1.	François Chollet, “Deep Learning with Python”,1st Edition, Manning Publisher, 2017.
2.	Aurélien Géron, “Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow”, 3rd Edition, O'Reilly, 2022.
References:	
1.	Sebastian Raschka, Vahid Mirjalili, “Python Machine Learning”, Packt Publishing, 2017.
2.	Online TensorFlow and PyTorch documentation
3.	Generative AI documentation for automated model tuning and interpretability

23AD906	GENAI ADVANCED PROMPT ENGINEERING & LLMS	1/0/4/3
Nature of Course	M (Practical Application)	
Prerequisites	Basic knowledge of Python programming and fundamentals of prompt engineering.	

Course Objectives:

1	To apply advanced prompt engineering techniques to craft effective prompts and optimize outputs from generative AI models.
2	To programmatically interact with large language models (LLMs) using Java libraries and APIs such as OpenAI and Hugging Face for AI-driven applications.
3	To design and implement API-driven workflows for tasks like content generation, summarization, and text analysis using GenAI services.
4	To address ethical considerations in generative AI, covering areas like bias mitigation, privacy protection, regulatory compliance, and fairness in AI-generated content.
5	To integrate generative AI functionalities into Java-based applications through foundational techniques, enabling the development of practical, AI-enhanced solutions.

Course Outcomes:

Upon completion of the course, students shall have ability to:

C906.1	Analyze advanced prompt engineering techniques to optimize GenAI outputs.	[AN]
C906.2	Implement programmatic interactions with LLMs using Java libraries and APIs (e.g., OpenAI, Hugging Face).	[AP]
C906.3	Design API-driven workflows for specific AI tasks such as content generation, summarization, and text analysis.	[AP]
C906.4	Analyze ethical practices in GenAI, including bias mitigation, privacy safeguards, adherence to regulatory frameworks, and ensuring fairness.	[AN]
C906.5	Apply foundational skills to integrate GenAI into practical applications, preparing for more advanced AI application-building courses.	[AP]

PROMPT ENGINEERING AND LLM INTEGRATION EXPLORING GENAI CAPABILITIES **5**

Introduction to Prompt Engineering: Basics of Prompt Engineering, Prompt Structure, Optimization Techniques, Complex Prompts, Common Pitfalls, Troubleshooting. Interacting with LLMs Programmatically: Java Libraries, API Keys, Request Handling, Response Processing, Key Functions, Error Handling, Rate Limits**.** Exploring GenAI Capabilities: Text Generation, Summarization, Sentiment Analysis, Text Classification, Language Translation, Use Cases, Output Quality.

API WORKFLOWS AND GENAI APPLICATIONS ETHICS, FAIRNESS, AND RESPONSIBLE AI **5**

API-Driven Workflows: API Workflows, Workflow Design, Automation Scripts, Workflow Efficiency, Model Integration, Security Considerations. Case Studies and Real-World Applications: GenAI Solutions, Customer Service, Education, Marketing, Domain-Specific Challenges, Effectiveness Evaluation. Ethics and Responsible AI: Biases in AI, Bias Mitigation, Privacy Safeguards, Regulatory Compliance, Fairness, Ethical Frameworks, Accountability Measures.

CAPSTONE PROJECT AND ASSESSMENT **5**

Project and Summative Assessment: Project Planning and Design, Solution Development, Testing and Debugging, Review and Refactoring, Project Documentation and Presentation, Summative Course-end Assessment.

Lab Experiments

Design an optimized few-shot prompt that generates creative product descriptions while minimizing hallucinations and ensuring factual accuracy.
Develop a Java script that programmatically interacts with an LLM API to generate contextual customer support responses.
Create and evaluate different variations of a chain-of-thought prompt for summarizing long-form research papers, ensuring conciseness without losing key insights. Compare the performance with a direct summarization prompt.
Implement an interactive chatbot using OpenAI API that refines user-generated prompts. Use the ReAct prompting method to enable reasoning and error correction before responding to the user.
Build an API-driven workflow in Java that automates blog content generation and stores the output in a database.
Develop an automation pipeline that takes raw user feedback, processes it using sentiment analysis, and categorizes responses for actionable insights. Use chain-of-thought reasoning to improve the classification of nuanced responses.
Integrate multiple AI models (e.g., OpenAI + Hugging Face) in a workflow to generate content followed by toxicity detection.
Implement a Java-based few-shot language translation tool using an LLM API and evaluate its performance across different dialects.
Develop an AI-driven resume screening tool that extracts and classifies key information from job applications.

Total Theory	**15 periods**
Total Lab	**60 periods**
Total	**75 periods**

Text Books:	
1	"You Look Like a Thing and I Love You" by Janelle Shane (for ethical perspectives).
2	OpenAI, Anthropic, or Cohere documentation for prompt engineering and API usage.
3	"Hands-On GenAI with Python" (online resources / platforms like GitHub & Medium tutorials).
Reference Books:	
1	OpenAI Cookbook — https://github.com/openai/openai-cookbook
2	Hugging Face API and Inference Documentation.
3	Ethical AI resources from Google, IBM, and Partnership on AI.
4	Flask and Fast API documentation for real-world web integration.

23AD901	DATA STORYTELLING AND VISUALIZATION	2/0/2/3
Nature of Course	M (Practical Application)	
Prerequisites	-	

Course Objectives:

1	To introduce learners to the fundamentals of data types, Excel functions, and spreadsheet operations.
2	To train learners in data cleaning, transformation, and preparation techniques using Excel tools.
3	To analyze data summarization using PivotTables and apply basic statistical concepts.
4	To understand in creating effective data visualizations to represent patterns, trends, and outliers.
5	To express the ability to craft compelling data narratives using dashboards and visual summaries.

Course Outcomes:

Upon completion of the course, students shall have ability to:

C901.1	Identify data categories and apply essential Excel functions to manipulate data efficiently	[U]
C901.2	Apply cleaning and transformation techniques to prepare datasets for analysis.	[AP]
C901.3	Generate summaries and insights using PivotTables and basic statistical techniques.	[AP]
C901.4	Evaluate meaningful visualizations for descriptive data analysis.	[AP]
C901.5	Design a real-world data storytelling project using dashboards and visual narratives based on industry-oriented practices.	[AP]

EXCEL DATA HANDLING AND FUNCTIONS **10**

Data Types and Functions – Data categories, structured vs. unstructured, XLOOKUP(), IF(), SUMIF(), SUMIFS, COUNTIF(), COUNTIFS, AVERAGEIF(), Logical Operators. Data Import and Cleanup – Import from text, remove blanks, trim, find and replace, remove duplicates.

ANALYSIS WITH PIVOTTABLES AND VISUAL TECHNIQUES **10**

PivotTables and Charts – Grouping, filtering, calculated fields, PivotCharts, Central Tendency Analysis – Mean, median, mode. Outlier Detection – Visual spotting, IQR, z-score basics - Bar and Column Charts – Categorical comparisons, stacked vs. clustered. Pie and Donut Charts – Composition representation. Histograms and Scatterplots – Distribution analysis, pattern recognition. Trend Lines and Custom Formatting – Forecasting, dynamic visuals.

DATA STORYTELLING, CORRELATION AND ASSESSMENTS **10**

Correlation and Spread – Variance, standard deviation, correlation coefficient. Storytelling with Data – Visual hierarchy, narrative building. Dashboard Design – Layout planning, dashboards, slicers, interactivity, interpretation - Apply storytelling techniques to a real-world dataset using Excel to design an interactive dashboard and narrative presentation.

List of Experiments

Classify different data types and apply functions like XLOOKUP, COUNTIF, and IF to perform logical operations for a given business scenario.
Import messy data from CSV/TEXT and clean it using trimming, find & replace, and removal of duplicates.
Create a PivotTable to summarize sales data by category, region, and time.
Generate a PivotChart to compare product performance across quarters.
Plot bar and pie charts using cleaned data to depict category-wise sales contribution.
Use histogram and scatter plot to analyze data spread and identify correlation.
Apply statistical formulas to calculate mean, median, mode, and detect outliers.
Design a multi-section Excel dashboard using slicers, conditional formatting, and charts.
Narrate a short data story integrating visuals and insights derived from an Excel analysis.
0. Apply data cleaning, pre-processing, visualization and storytelling techniques to a real-world dataset using Excel to design an interactive dashboard.

Total Theory Total	**30 periods**
Lab	**30 periods**
Total	**60 periods**

Text Books:	
1	"Data Visualization with Excel Dashboards and Reports" — Dick Kusleika.
2	"Storytelling with Data: A Data Visualization Guide for Business Professionals" — Cole Nussbaumer Knaflic.
Reference Books:	
1	Microsoft Excel Online Documentation.
2	Blogs and Tutorials on Excel PivotTables, Charts, and Dashboards.
3	Data storytelling case studies and online courses (LinkedIn Learning, Coursera).

23AD903	PROBLEM SOLVING WITH ANALYTICAL & DESIGN THINKING	2/0/2/3
Nature of Course	F (Theory Programming)	
Prerequisites	Nil	

Course Objectives:

1	To identify and deconstruct real-world problems through root cause analysis and user-centric research.
2	To generate innovative ideas using structured ideation techniques and frame actionable problem statements.
3	To organize, prioritize, and visualize ideas through the creation of rapid prototypes and user flow mockups.
4	To explore solutions with users and refine designs through iterative feedback and continuous improvement.
5	To design and present real-world solutions and plan further improvements through collaborative decision-making.

Course Outcomes:

Upon completion of the course, students shall have ability to:

C903.1	Identify root causes of real-world problems and build user empathy through research tools and techniques.	[AN]
C903.2	Generate multiple creative solutions and translate user needs into actionable problem statements.	[AN]
C903.3	Organize and prioritise ideas using mapping tools and create low-fidelity prototypes.	[AN]
C903.4	Conduct usability testing and refine designs based on structured user feedback and observation.	[AN]
C903.5	Apply analytical and design thinking practices in an end-to-end industry-oriented problem-solving project.	[AP]

PROBLEM DISCOVERY AND RESEARCH **10**

Problem Identification & Understanding – Spot surface-level vs deeper issues – Apply 5 Whys and Fishbone tools – Break down complex challenges – Frame a clear problem statement. Understanding the User's World – Create Empathy Maps – Draft User Journey Maps – Identify pain points and behavior patterns – Use journey maps to define the real problem. Defining the Problem Statement – Synthesise key user insights – Frame a Point of View (POV) – Write focused "How Might We" questions – Align problem with user/business needs.

IDEATION, PROTOTYPING AND VISUALISATION **10**

Generating Potential Solutions – Run Crazy 8s sketch activity – Use SCAMPER technique Build on team ideas – Avoid early filtering or judging. Prioritising & Organising Ideas – Cluster ideas using Affinity Mapping – Build early process flows – Use Impact–Effort matrix – Select one idea to pursue. Translate ideas into quick draft – Sketch step-by-step user flows – Draw simple screen layouts – Build quick paper-based mockups – Walk others through the concept.

TESTING, REFINEMENT AND FINAL PROJECT EXECUTION **10**

Testing with Users – Conduct usability tests – Use Think-Aloud Protocol – Observe and take silent notes – Identify confusion or friction points. Refine designs using user feedback – Identify what to improve from feedback – Prioritise using Impact–Effort mapping – Document decisions in a changelog – Refine prototypes. Final Project Execution – Project and Summative Assessment involving real-world problem definition, user research, ideation, prototyping, testing, and final presentation.

List of Experiments

Root Cause Analysis: Use 5 Whys and Fishbone diagram to identify the core of a given problem.
Empathy Mapping: Create empathy maps and journey maps based on user interviews or personas.
Problem Framing: Write Point of View (POV) statements and corresponding "How Might We" questions.
Idea Generation: Conduct a Crazy 8s activity to generate 8 creative solutions in 8 minutes.
SCAMPER Exercise: Apply SCAMPER technique to improve an existing product or service.
Idea Prioritisation: Use Affinity Mapping and Impact–Effort Matrix to select a feasible solution.
Prototyping: Build a low-fidelity paper prototype and prepare a walk-through of the concept.
User Testing: Conduct Think-Aloud usability testing with 2–3 users and document observations.
Iteration Log: Maintain a changelog to record design decisions and refinements based on feedback.
0. Mini Project: Execute a mini end-to-end design sprint (Empathize → Define → Ideate → Prototype → Test) on a campus or community problem.

Total Theory	**30 periods**
Total Lab	**30 periods**
Total	**60 periods**

Text Books:	
1	Tim Brown, Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation, Harper Business.
2	Jeanne Liedtka, Design Thinking for the Greater Good: Innovation in the Social Sector, Columbia Business School Publishing.
Reference Books:	
1	IDEO.org, The Field Guide to Human-Centered Design.
2	Kelley, T., The Art of Innovation.
3	Selected online case studies and design thinking toolkits (IDEO, Stanford d.school).

23AD904	POWER BI	2/0/2/3
Nature of Course	M (Practical Application)	
Prerequisites	-	

Course Objectives:

1	To identify the complete workflow of extracting, cleaning, and transforming data using Power Query.
2	To gain knowledge in preparing structured datasets and designing efficient data models for scalable analytics.
3	To understand the use of DAX expressions for aggregations, measures, and implementing business logic.
4	To be familiar with the process of designing interactive dashboards and applying effective data storytelling techniques.
5	To analyse AI-powered analytics features and the secure publishing and management of Power BI reports for collaboration.

Course Outcomes:

Upon completion of the course, students shall have ability to:

C904.1	Extract, clean, and transform raw datasets using Power Query and ETL features in Power BI.	[AN]
C904.2	Structure data and implement robust data models for performance and scalability.	[AN]
C904.3	Use DAX to build dynamic metrics, perform business calculations, and derive insights.	[AP]
C904.4	Design compelling dashboards with interactive visuals for effective data storytelling.	[C]
C904.5	Apply advanced DAX and AI analytics to generate predictive insights and deliver a course-end project simulating real business reporting.	[AP]

DATA PREPARATION AND TRANSFORMATION **10**

Data Extraction and Transformation - Power Query Editor- ETL techniques - connecting to data sources - cleansing and shaping data. Data Preparation and Structuring - Handling missing data - data types - appending and merging queries - data formatting best practices. Data Modeling and Optimization - Creating Relationships - star and snowflakes schema design - calculated columns – role playing dimensions.

MODELING AND DAX CALCULATIONS **10**

Model Optimization - Relationship management - model size reduction - use of aggregations - performance tuning. Basic DAX Calculations – Syntax - calculated columns vs measures – aggregations- conditional – logical - time intelligence functions. Advanced DAX - Context transition, CALCULATE -FILTER, scenario-based analysis, KPIs.

REPORTING, VISUALIZATION AND PROJECT EXECUTION **10**

Data Visualization and Storytelling – Chart selection - visual formatting – narrative driven dashboards – drill through - bookmarks. Interactive Reports – Slicers – filters – tooltips – buttons - custom visuals - cross-filtering. AI-Powered Analytics - Q&A visuals - Smart Narrative - Decomposition Tree - Key Influencers. Project and Summative Assessment including DAX model building - interactive dashboard creation - AI insights - and secure report sharing.

LIST OF LAB EXPERIMENTS

Data Cleaning in Power Query: Load a messy CSV and perform basic cleaning such as removing nulls, renaming columns, and filtering rows.
ETL Workflow: Connect multiple data sources (CSV, Excel), perform merge and append operations in Power Query.
Model Creation: Create fact and dimension tables from the dataset and establish relationships for a sales dashboard.
Calculated Columns and Measures: Use DAX to create calculated columns (e.g., Profit = Sales - Cost) and dynamic measures (e.g., Year-over-Year Growth).
Time Intelligence: Create Month-to-Date and Year-to-Date measures using DAX time intelligence functions.
Dashboard Design: Develop a sales performance dashboard using line charts, KPI cards, slicers, and bar charts.
Drill-through and Bookmarks: Add page navigation and drill-through capabilities to a dashboard with customized tooltips.
AI Insights Integration: Use decomposition tree and key influencers visuals for AI-assisted insights.
Secure Report Publishing: Publish a report to Power BI Service, set access controls, and schedule data refresh.
). Mini Capstone Project: Execute a complete reporting cycle including data cleaning, modelling, DAX, visualization, and secure publishing for a retail business use case.

Total Theory	**30 periods**
Total Lab	**30 periods**
Total	**60 Periods**

Text Books:	
1	Alberto Ferrari & Marco Russo, The Definitive Guide to DAX: Business intelligence with Microsoft Excel, SQL Server Analysis Services, and Power BI, Microsoft Press.
2	Daniil Maslyuk, Mastering Microsoft Power BI, Packt Publishing.
Reference Books:	
1	Reza Rad, Power BI from Rookie to Rock Star, RADACAD.
2	Ruth Pozuelo Martinez, Learning Power BI, Curbal.
3	Official Microsoft Power BI Documentation and Community resources.

23CS901	IMPLEMENTING AND ADMINISTERING ENTERPRISE NETWORKS	2/0/2/3
Nature of Course	D (Theory Application)	
Pre-requisite(s): -		

Course Objectives:

1	To introduce the fundamentals of network design using OSI and TCP/IP models for various networking scenarios.
2	To develop logical networking skills for small to large-scale networks including LAN, MAN, and WAN environments.
3	To train learners to configure routers and switches using Cisco's Internetworking Operating System (IOS).
4	To equip students with skills in network traffic monitoring, analysis, and troubleshooting using industry-grade tools.
5	To familiarize learners with the use of GenAI tools in detecting network anomalies and strengthening network security.

Course Outcomes:

Upon completion of the course, students shall have ability to

C901.1	Analyse communication protocols using OSI and TCP/IP models and apply them to design efficient networks.	[AN]
C901.2	Design and implement logical networks for LAN, MAN, and WAN using simulation and configuration tools.	[AP]
C901.3	Configure routers and switches using Cisco IOS to establish and secure reliable network connectivity.	[AP]
C901.4	Troubleshoot and optimize network performance using techniques like subnetting, partitioning, and IP planning.	[AN]
C901.5	Build an end-to-end network project integrating GenAI-based anomaly detection with tools like Wireshark and SNMP, adhering to industry practices.	[AP]

NETWORK FOUNDATIONS AND LOGICAL NETWORK DESIGN **10**

OSI and TCP/IP models, data encapsulation and de-encapsulation, Ethernet standards, types of cables and connectors, network interface devices, switching basics, collision and broadcast domains, LAN, MAN, WAN architectures, IP addressing and subnetting, VLSM and CIDR, network partitioning, logical topology planning, VLAN creation and configuration, trunking and inter-VLAN routing.

ROUTING AND SWITCHING **10**

Router and switch boot process, Cisco IOS command-line interface, static routing, RIP and OSPF configuration, routing table management, switch configuration, MAC address tables, port security settings,

NETWORK ANALYSIS & ANOMALY DETECTION **10**

Traffic sniffing using Wireshark, protocol and packet analysis, identifying latency and packet loss, SNMP-based monitoring with Zabbix, event correlation, GenAI-based anomaly detection, alert generation and reporting. End-to-end network design and simulation, router and switch configuration, traffic monitoring using Wireshark, integration of GenAI tools for anomaly detection, documentation and reporting of findings.

TOTAL PERIODS: 30

Lab Component:

1	Simulate OSI layer communication using Packet Tracer and analyze encapsulation and de-encapsulation at each layer.
2	Design a LAN topology in GNS3 and assign static IPs to configure connectivity among nodes.
3	Perform subnetting and configure appropriate addressing schemes for segmented networks.
4	Create and configure VLANs and enable inter-VLAN routing in a switch-based topology.
5	Configure and verify static routing and RIP protocol on multiple routers.
6	Use Wireshark to capture live packets, filter protocols like ICMP, HTTP, DNS, and analyze headers.

7	Diagnose network issues using tools like ping, traceroute, and show commands in Cisco IOS.	
8	Monitor network performance using Zabbix with SNMP configurations to track CPU, memory, and interface traffic.	
9	Detect and classify network anomalies using GenAI prompts applied to Wireshark data logs.	
10	Complete an integrated project to design, configure, monitor, and report on a secure and efficient network using simulation and GenAI tools.	
	TOTAL (LAB) PERIODS	**30**
	TOTAL PERIODS	**60**

TEXT BOOKS:

1	Behrouz A. Forouzan, “Data Communications and Networking”, 5th Edition, McGraw-Hill Education, 2017

REFERENCE BOOKS:

1	Wayne Lewis, “Cisco Certified Network Associate (CCNA) Routing and Switching 200-120 Official Cert Guide Library”, Cisco Press, 2013
2	Richard Deal, “CCNA Routing and Switching Portable Command Guide”, 4th Edition, Cisco Press, 2016
3	Chris Sanders, “Practical Packet Analysis: Using Wireshark to Solve Real-World Network Problems”, 3rd Edition, No Starch Press, 2007
4	William Stallings, “Network Security Essentials: Applications and Standards”, 6th Edition, Pearson, 2016

23CS902	LINUX SYSTEM ADMINISTRATION	20/2/3
Nature of Course	D (Theory Application)	
Pre-requisite(s): -		

Course Objectives:

1	To introduce learners to Linux system architecture, essential commands, and shell environment.
2	To equip students with skills to install Linux distributions, manage partitions, and configure file systems.
3	To enable management of users, groups, permissions, services, and disk operations.
4	To teach automation of administrative tasks using Bash scripting and GenAI tools.
5	To train learners to configure network services, implement security hardening, and centralized monitoring in a Linux environment.

Course Outcomes:

Upon completion of the course, students shall have ability to

C902.1	Install and configure Linux systems with manual disk partitioning and filesystem setup.	[AP]
C902.2	Manage users, groups, files, permissions, and services for secure system operations.	[AN]
C902.3	Troubleshoot Linux services and processes to maintain availability and performance.	[AN]
C902.4	Automate administrative tasks using Bash scripting and integrate GenAI tools effectively.	[AN]
C902.5	Complete a capstone project applying Linux administration concepts and industry practices in a simulated environment.	[AN]

LINUX INSTALLATION AND USER MANAGEMENT **10**

Linux installation using ISO files, manual disk partitioning, Linux boot process, file system structure and types, working with mount and unmount commands, terminal usage and basic Linux commands, package management in Debian and RedHat systems, User and group management, setting file and directory permissions using chmod and chown, managing access control lists (ACL), disk partitioning using fdisk and parted, managing Linux processes and services, service status monitoring using systemctl and journalctl.

AUTOMATION AND SHELL SCRIPTING **10**

Writing and debugging Bash scripts, automating routine tasks with cron and at, managing process priorities using nice and renice, multitasking in shell, system update and upgrade operations, using GenAI tools to generate Linux administrative scripts,

NETWORKING AND MONITORING SERVICES **10**

Configuring network interfaces and services, using nmtui and Network Manager, implementing DHCP and DNS services, enabling remote access with SSH and RDP, centralized logging with rsyslog and syslog-ng, log management using logrotate, SNMP-based monitoring using Zabbix.Capstone project integrating Linux installation, configuration, service management, scripting, and monitoring tasks, demonstrating real-world system administration and automation practices using VirtualBox, GenAI, and remote tools.

TOTAL PERIODS: 30

Lab Component:

1	Install Ubuntu or CentOS with manual partitioning using VirtualBox.
2	Create, modify, and delete users and groups, and enforce password aging policies.
3	Configure and test file and directory permissions using chmod, chown, and ACL.
4	Partition disks and create file systems using fdisk, mkfs, and mount them manually.
5	Monitor and restart services using systemctl and diagnose logs using journalctl.
6	Write a Bash script to automate the creation of users and configuration of a firewall.
7	Schedule automated backup tasks using cron and implement basic recovery with rsync.
8	Configure static IPs, test DHCP and DNS settings, and access systems remotely via SSH.

9	Implement centralized logging using rsyslog and rotate logs with logrotate.	
10	Generate and validate Linux admin scripts using GenAI for real-time task automation.	
	TOTAL (LAB) PERIODS	**30**
	TOTAL PERIODS	**60**

TEXT BOOKS:

1	Tommy Singleton, “Linux Administration Handbook”, 2nd Edition, Pearson Education, 2006

REFERENCE BOOKS:

1	Evi Nemeth, Garth Snyder, Trent R. Hein, Ben Whaley, Dan Mackin, “UNIX and Linux System Administration Handbook”, 5th Edition, Pearson, 2017
2	William E. Shotts Jr., “The Linux Command Line: A Complete Introduction”, No Starch Press, 2012.
3	Mark G. Sobell, “A Practical Guide to Linux Commands, Editors, and Shell Programming”, 4th Edition, Pearson, 2017
4	Nemeth, Hein, Snyder, Whaley, “Linux Administration: A Beginner’s Guide”, 8th Edition, McGraw-Hill Education, 2020

23CS903	INFORMATION SECURITY SYSTEMS	2/0/2/3
Nature of Course	D (Theory Application)	
Pre-requisite(s): -		

Course Objectives:

1	To introduce learners to the foundational principles and core objectives of information security, including the CIA triad, asset protection, and risk assessment methodologies.
2	To train students to implement secure authentication and enforce access control mechanisms in enterprise systems.
3	To equip learners with knowledge and tools to build secure network perimeters using firewalls, IDS/IPS systems, and encrypted communication protocols.
4	To enable students to understand and apply cryptographic techniques for secure data exchange and identity verification.
5	To leverage GenAI tools to enhance threat detection, malware analysis, and develop real-world security responses.

Course Outcomes:

Upon completion of the course, students shall have ability to

C903.1	Explain the CIA triad, conduct risk assessments, and align security implementations with industry standards.	[U]
C903.2	Design and enforce authentication policies using passwords, hashing techniques, and multi-factor authentication.	[AP]
C903.3	Configure and manage network security defences such as firewalls, IDS/IPS, and VPNs to prevent external threats.	[AN]
C903.4	Apply symmetric and asymmetric cryptography, generate keys, and secure data using encryption, digital signatures, and PKI.	[AP]
C903.5	Use GenAI and open-source tools to analyse and mitigate malware threats and implement security controls in practical scenarios.	[AP]

FUNDAMENTALS OF INFORMATION SECURITY **10**

CIA triad concepts, assets and vulnerabilities, threats and attacks, security goals and objectives, risk analysis methods, information security frameworks and compliance standards (ISO 27001, NIST), security governance and security policiesm,Password policy enforcement, hashing and salting, multi-factor authentication methods, access control models (MAC, DAC, RBAC), implementation of IAM (Identity and Access Management), directory services, SSO integration.

NETWORK PERIMETER PROTECTION **10**

Firewall types and configuration (stateful, stateless), VPN setup and tunneling protocols, IDS/IPS concepts and implementation (Snort, Suricata), secure network protocol usage (SSH, HTTPS, SFTP), transport layer security, port scanning and hardening,

CRYPTOGRAPHY AND DATA SECURITY **10**

Symmetric and asymmetric encryption techniques (AES, RSA), hashing algorithms (SHA, MD5), Kerberos, LDAP, digital signature generation and verification, public key infrastructure (PKI), usage of OpenSSL for encryption and signing, certificate generation and validation.

Capstone project on secure system design including cryptographic implementation, firewall and VPN setup, malware traffic detection using GenAI-based tools. Summative evaluation through documentation, testing, and practical defense strategy presentation.

TOTAL PERIODS: 30

Lab Component:

1	Analyse and document security gaps in a given system using the CIA triad model.
2	Implement password hashing using SHA256 and validate salted hash results.
3	Set up multi-factor authentication using tools like Google Authenticator or DUO.
4	Simulate RBAC (Role-Based Access Control) in a Linux or web application environment.
5	Configure iptables or pfSense to create firewall rules and log denied packets.

6	Capture and analyse potentially malicious traffic using Wireshark filters.	
7	Encrypt and decrypt sensitive files using OpenSSL command-line utilities.	
8	Create and use public/private key pairs for signing and verification tasks.	
9	Use a GenAI-based tool to simulate malware detection and analyse log patterns.	
10	Final integrated project to demonstrate understanding of access control, cryptographic security, and perimeter defence mechanisms.	
	TOTAL (LAB) PERIODS	**30**
	TOTAL PERIODS	**60**

TEXT BOOKS:

1	William Stallings, “Cryptography and Network Security: Principles and Practice”, 8th Edition, Pearson Education, 2023
2	Michael E. Whitman and Herbert J. Mattord, “Principles of Information Security”, 7th Edition, Cengage Learning, 2023

REFERENCE BOOKS:

1	Behrouz A. Forouzan, “Cryptography and Network Security”, McGraw Hill Education, 2007
2	Mark Rhodes-Ousley, “Information Security: The Complete Reference”, 2nd Edition, McGraw Hill Education, 2012
3	Nitesh Dhanjani, “Network Security Tools: Writing, Hacking, and Modifying Security Tools”, O’Reilly Media, 2005
4	Shon Harris and Fernando Maymi, “CISSP All-in-One Exam Guide”, 8th Edition, McGraw Hill Education, 2019

23CS904	LOW-CODE NO-CODE APPLICATION BUILDING	2/0/2/3
Nature of Course	D (Theory Application)	
Pre-requisite(s): -		

Course Objectives:

1	To grasp foundational concepts such as programming logic techniques, databases, user interface elements, and basic application logic to create robust LCNC solutions.
2	To get familiar with LCNC tools, HTML5/CSS3, ChatGPT, and discover their applications in streamlining development.
3	To create form-based applications with customized user interfaces for data collection and analysis.
4	To build workflow-driven applications that optimize business processes using automation triggers and logic.
5	To design and customize reports within LCNC applications to effectively visualize and communicate data insights.

Course Outcomes:

Upon completion of the course, students shall have ability to

C904.1	Explain LCNC development principles, identify use cases, and select suitable tools and techniques.	[AN]
C904.2	Design user interfaces and develop applications using drag-and-drop components and responsive layout techniques.	[AP]
C904.3	Develop dynamic form-based applications with validation, conditional logic, and data collection.	[AN]
C904.4	Integrate APIs and configure workflow automation using triggers and business logic.	[AN]
C904.5	Visualize application data using built-in reporting tools and finalize a comprehensive project solution.	[AN]

FOUNDATIONS OF LCNC AND UI DESIGN **10**

Low-code no-code principles, citizen development and its role in business, benefits and limitations of LCNC, programming logic concepts, working with databases and user interface elements, overview of LCNC platforms, basic HTML5 and CSS3 for customization, ChatGPT for assistance in development, API integration basics, fundamentals of user-centered design, creating responsive interfaces, visual hierarchy and usability principles,

WORKFLOW AUTOMATION AND DATA VISUALIZATION **10**

Form design using LCNC platforms, dynamic fields and conditional logic, multi-step forms and layout optimization, input validation and error handling, designing application logic, process automation through workflows, creating and managing custom triggers, integrating external services into workflows, debugging and optimizing automated processes. Designing and embedding interactive dashboards, choosing effective visualization types, working with real-time data sources, report creation and layout customization, grouping and filtering data, using built-in connectors for cloud services, configuring custom APIs for data fetch and sync, troubleshooting integration issues, testing system interoperability and performance,

ROLE-BASED ACCESS AND SECURITY PRACTICES **10**

User authentication techniques, configuring user roles and access permissions, implementing secure data handling and validation, best practices in managing application sessions and credentials, audit logging and activity tracking, protecting APIs and third-party integrations, enforcing secure deployment guidelines across LCNC applications.

Final capstone project integrating UI design, form development, workflow automation, API integration, and reporting, project planning and scope definition, iterative development and testing, debugging and error resolution, final deployment and presentation, course-end summative assessment and individual reflection.

TOTAL PERIODS: 30

Lab Component:

1	Create a simple LCNC-based application to collect user data using form components.
2	Design a responsive user interface layout with text fields, dropdowns, and buttons.
3	Implement form validation rules and configure multi-step navigation between form sections.

4	Automate a workflow that sends notifications based on form submissions using triggers.	
5	Create dynamic dashboards with charts to visualize collected data in real-time.	
6	Integrate a third-party API into your application and display external data in a table.	
7	Build and export a custom report with filtered and grouped data insights.	
8	Use a GenAI tool like ChatGPT to generate field logic and UI text recommendations.	
9	Configure user roles, permissions, and access control for multi-level users.	
10	Develop a capstone LCNC project integrating forms, workflows, API calls, and reporting features.	
	TOTAL (LAB) PERIODS	**30**
	TOTAL PERIODS	**60**

TEXT BOOKS:	
1	Lim Mei Ying, Sriram Krishnan, “Low-Code Application Development with Appian”, 1st Edition, Packt Publishing, 2022.
2	Mark Goodyear, “HTML5 and CSS3: Visual QuickStart Guide”, Peachpit Press, 2011.
REFERENCE BOOKS:	
1	Jan Vanthienen and Monique Snoeck, “Low-Code Development: A Practical Guide for Business Transformation”, Springer, 2021
2	Matt Cavanagh, “Building Low-Code Applications with Mendix”, 1st Edition, Packt Publishing, 2021.
3	Jon Duckett, “HTML and CSS: Design and Build Websites”, 1st Edition, Wiley, 2011.
4	George Reese, “Cloud Application Architectures: Building Applications and Infrastructure in the Cloud”, O’Reilly Media, 2009
5	Vijay Kumar Velu, Robert Beggs, “Mastering Kali Linux for Advanced Penetration Testing”, 3rd Edition, Packt Publishing, 2019

23CS905	VIRTUALIZATION, CLOUD COMPUTING & SYSOPS	1/0/4/3

Nature of the Course: F (Theory Programming)

Pre-requisite(s): -

Course Objectives:

1	To understand core principles of IT virtualization, containerization, and cloud computing.
2	To learn to implement and manage virtualized infrastructure and storage systems.
3	To use Docker containers for efficient application deployment.
4	To explore AWS cloud services and architectures for scalable and resilient deployments.
5	To integrate GenAI tools for enhancing security and cloud operations.
6	To design and implement fault-tolerant cloud solutions with real-time use cases.

Course Outcomes:

Upon completion of the course, students shall have ability to

C905.1	Understand the foundational concepts of virtualization, containerization, and cloud deployment models.	[U]
C905.2	Implement and manage virtual machines, storage, and networking using hypervisors and cloud-based tools.	[AP]
C905.3	Explore containerization using Docker and distinguish it from traditional virtualization approaches.	[AN]
C905.4	Design secure and resilient cloud architectures using GenAI tools and AWS services.	[AP]
C905.5	Gain proficiency in AWS SysOps tasks including resource provisioning, automation, and infrastructure monitoring.	[AP]

VIRTUALIZATION AND CLOUD FOUNDATIONS **5**

Capital expenditure vs operational expenditure in IT, cloud deployment patterns and service models (IaaS, PaaS, SaaS), just-in-time provisioning concepts, hypervisor types and architecture, creating and managing virtual machines using VirtualBox, introduction to virtual storage and networking, types of virtual networks and configurations.

CONTAINERIZATION AND INFRASTRUCTURE MANAGEMENT **5**

Containerization fundamentals, how Docker differs from hypervisors, building Docker images with Dockerfile, running and managing Docker containers, creating and managing Docker networks, use cases for containerized workloads, integrating containers in cloud environments. Utilizing GenAI tools for security analysis and compliance, introduction to AWS cloud infrastructure, regions and availability zones, AWS identity and access management (IAM), managing user accounts and billing with AWS Organizations, designing and implementing secure Virtual Private Clouds (VPC), configuring access control and network segmentation, introduction to CloudTrail for auditing.

SYSOPS ON AWS AND SCALABLE DEPLOYMENTS **5**

Provisioning applications on AWS using EC2, managing block storage with EBS, hosting static data using S3, working with RDS for relational databases, configuring auto-scaling and load balancing, monitoring services with AWS tools, deploying fault-tolerant architectures using AWS services, hands-on experience with real-time deployment and infrastructure automation.

Final project focused on designing, implementing, and deploying a secure and scalable cloud-native application, infrastructure planning using VirtualBox and AWS, integration of Docker containers, applying GenAI tools for security evaluation, monitoring and optimization, project documentation, presentation, and summative evaluation.

TOTAL PERIODS: 15

Lab Component:

1	Install and configure VirtualBox to create and manage multiple virtual machines.
2	Design and deploy a virtual network using host-only and bridged network adapters in VirtualBox.
3	Set up and test Docker containers by creating a Dockerfile and launching multi-container applications.
4	Create custom Docker networks and connect multiple containers for inter-container communication.

5	Provision and configure an EC2 instance on AWS and access it using SSH.
6	Create and attach Elastic Block Store (EBS) volumes to EC2 instances and perform mount operations.
7	Design and deploy a Virtual Private Cloud (VPC) with public and private subnets using AWS.
8	Use AWS IAM to create users, groups, and policies, and test access control scenarios.
9	Enable and analyze CloudTrail logs for auditing AWS account activities.
10	Plan and execute a project to deploy a secure, fault-tolerant web application using EC2, S3, RDS, and Docker containers.

TOTAL (LAB) PERIODS	**60**
TOTAL PERIODS	**75**

TEXT BOOKS:

1	Tom Clark , “Designing and Implementing Linux Virtualization”, Packt Publishing,
2	Karl Matthias, Sean P. Kane ,”Docker: Up & Running: Shipping Reliable Containers in Production”, 3rd Edition O’Reilly Media, 2023.
3	Michael Wittig, Andreas Wittig, “Amazon Web Services in Action”, 3rd Edition Manning Publications, 2023.

REFERENCE BOOKS:

1	James Turnbull, “The Docker Book: Containerization is the New Virtualization”, Turnbull Press, 1st Edition, 2019.
2	Eric Ligman, “AWS Certified SysOps Administrator Official Study Guide”, Wiley, 2017.
3	Thomas Erl, “Cloud Computing: Concepts, Technology & Architecture”, Prentice Hall, 2013.
4	Stuart Scott, “AWS Certified Solutions Architect Study Guide: Associate (SAA-C03) Exam”, 4th Edition, Wiley, 2022.
5	Benjamin Caudill and Karl Gilbert, “Hands-On AWS Penetration Testing with Kali Linux”, Packt Publishing Limited, 2019.

23CS906	CONTINUOUS MONITORING AND OBSERVABILITY	1/0/4/3
Nature of Course	F (Theory Programming)	
Pre-requisite(s): -		

Course Objectives:

1	To understand the need for configuration management and infrastructure as code in cloud environments.
2	To learn to automate server configurations using Ansible and manage resources through AWS CloudFormation.
3	To apply observability principles to monitor AWS infrastructure using CloudWatch metrics, events, and alarms.
4	To configure monitoring dashboards and set up alerting mechanisms for proactive cloud operations.
5	To analyse AWS logs to troubleshoot, detect issues, and implement response strategies using GenAI tools.

Course Outcomes:

Upon completion of the course, students shall have ability to

C906.1	Understand the fundamentals of Configuration Management and Infrastructure as Code using Ansible and AWS CloudFormation.	[U]
C906.2	Apply Ansible to automate configuration tasks and manage cloud inventory using playbooks, variables, and modules.	[AP]
C906.3	Develop proficiency in designing and deploying AWS infrastructure using CloudFormation templates and stacks.	[AN]
C906.4	Monitor AWS infrastructure by setting up CloudWatch dashboards, custom metrics, events, and alarms.	[AN]
C906.5	Analyse AWS logs to detect patterns, optimize performance, and ensure system health.	[AN]

CONFIGURATION MANAGEMENT WITH ANSIBLE **5**

Introduction to Configuration Management, Introduction to Ansible, Defining Inventory Files, Ansible Playbooks, Group and Host Variables, Ansible Modules and Galaxy,

INFRASTRUCTURE AUTOMATION AND MONITORING AWS INFRASTRUCTURE **5**

AWS CloudFormation Overview, Writing YAML Templates, Defining Resources, Stack Management, Template Validation, Updating and Deleting Stacks, Linting Best Practices. Introduction to Observability, AWS CloudWatch Metrics, Creating Dashboards, Monitoring EC2, EBS, RDS, ELB, Creating Custom Metrics, Common Graphs and Visualizations.

CLOUDWATCH EVENTS, LOGS, AND ALERTS **5**

CloudWatch Events and Triggers, setting up Alarms, Using AWS Logs for Troubleshooting, Log Insights, Automating Alert Responses, Using GenAI for Alert Analysis

Real-world implementation of monitoring solution using Ansible and CloudFormation, with complete observability via CloudWatch. Includes alert configuration and log analysis using GenAI.

TOTAL PERIODS: 15

Lab Component:

1	Install and configure Ansible on a Linux system and define a basic inventory file.
2	Write and execute an Ansible playbook to configure Apache web server on EC2 instances.
3	Use Ansible variables and handlers to customize service deployment and restart services on change.
4	Create a CloudFormation template to deploy an EC2 instance and attach an EBS volume.
5	Validate and deploy a CloudFormation stack and update it to reflect infrastructure changes.
6	Set up CloudWatch to monitor EC2 instance metrics and create a custom dashboard.
7	Configure CloudWatch to collect and display metrics for EBS, RDS, and ELB services.
8	Create a CloudWatch alarm to notify when CPU utilization exceeds a defined threshold.

9	Analyse AWS CloudTrail logs to track API activities and detect unauthorized changes.	
10	Simulate and respond to an infrastructure alert using automation and observability best practices.	
	TOTAL (LAB) PERIODS	**60**
	TOTAL PERIODS	**75**

TEXT BOOKS:

1	Jeff Geerling, “Ansible for DevOps: Server and Configuration Management for Humans”, 1st Edition, Midwestern Mac, 2015.
2	Michael Wittig, Andreas Wittig, “Amazon Web Services in Action”, 3rd Edition, Manning Publications, 2023.

REFERENCE BOOKS:

1	Lorin Hochstein, “Ansible: Up and Running — Automating Configuration Management and Deployment the Easy Way”,3rd Edition, O’Reilly Media, 2022.
2	Kief Morris, “Infrastructure as Code: Dynamic Systems for the Cloud Age”, 2nd Edition, O’Reilly Media, 2020.
3	Eric Ligman, “AWS Certified SysOps Administrator Official Study Guide”, 3rd Edition, Wiley, 2024.
4	Mike Julian, “Practical Monitoring: Effective Strategies for the Real World”, 1st Edition, O’Reilly Media, 2017.

23CY901	CLOUD COMPUTING AND CONTAINERIZED VIRTUAL INFRASTRUCTURE	2/0/2/3
Nature of Course	D (Theory Application)	
Pre-requisite(s): -		
Course Objectives:		
1	To understand the fundamental concepts of cloud computing, virtualization, and containerization.	
2	To learn to deploy and manage virtual machines using hypervisor technologies.	
3	To explore virtual networking and storage concepts in a cloud-based environment.	
4	To implement containerization using Docker for lightweight and scalable workload management.	
5	To apply GenAI tools to enhance cloud infrastructure security and risk analysis	
Course Outcomes: Upon completion of the course, students shall have ability to		
C901.1	Explain cloud computing models, virtualization techniques, and infrastructure provisioning.	[U]
C901.2	Deploy and manage virtual machines using tools like VirtualBox.	[AP]
C901.3	Configure and manage virtual storage and networking systems.	[AP]
C901.4	Build and manage containerized applications using Docker.	[AP]
C901.5	Secure containerized and virtualized environments using GenAI-powered tools.	[AP]

CLOUD COMPUTING AND VIRTUALIZATION CONCEPTS **10**

Capital Expenditure vs. Operational Expenditure, Cloud Deployment Patterns, Service Models (IaaS, PaaS, SaaS), Just-in-Time Provisioning, Auto-scaling and Elasticity, Types of Hypervisors and their use cases

VIRTUAL MACHINE, STORAGE MANAGEMENT AND DOCKER CONTAINERIZATION **10**

Creating Virtual Machines using Oracle VirtualBox, Snapshot and Cloning, Resource Allocation and Optimization, Virtual Storage Concepts (Local and Remote), Creating and Managing Virtual Disks in Linux. Types of Virtual Networking (NAT, Bridged, Host-Only), IP Configuration and Access, Containerization with Docker, Docker vs. Hypervisors, Creating Docker Images with Dockerfile, Running and Networking Docker Containers,

GENAI-DRIVEN CLOUD SECURITY **10**

Security Risks in Virtualized Infrastructure, Cloud Security Principles, Role-Based Access Control, Introduction to GenAI Tools for Threat Detection, GenAI-based Cloud Security Automation and Best Practices. Plan and implement a secure virtual and containerized cloud infrastructure; include hypervisors, Docker, and GenAI security components. Activities include architecture design, deployment, testing, documentation, and presentation.

TOTAL PERIODS:	**30**

Lab Component:

1	Set up a virtual machine in Oracle VirtualBox and install a Linux OS.
2	Simulate just-in-time provisioning and auto-scaling using cloud deployment tools.
3	Configure virtual networks (NAT and bridged) and verify VM-to-VM connectivity.
4	Attach and partition virtual storage in a Linux-based virtual machine.
5	Create a Dockerfile and build a custom container image.
6	Deploy multiple Docker containers and link them via Docker networks.
7	Set up port forwarding in Docker and expose containerized services.
8	Compare and analyze resource usage between VMs and Docker containers.
9	Use a GenAI tool to identify vulnerabilities in a containerized setup.
10	Design and present a project on secure cloud infrastructure using virtualization and containers
TOTAL (LAB) PERIODS	**30**
TOTAL PERIODS	**60**

TEXT BOOKS:	
1	Virtualization Essentials — Matthew Portnoy, Wiley, 2nd Edition.
2	Docker: Up & Running — Shipping Reliable Containers in Production — Karl Matthias, Sean P. Kane, O'Reilly Media, 3rd Edition
REFERENCE BOOKS:	
1	Mastering Virtual Machine Manager 2019 — Sarmad Ali, Packt Publishing, Latest Edition.
2	The Docker Book: Containerization is the New Virtualization — James Turnbull, Turnbull Press, Latest Edition.
3	AWS Certified SysOps Administrator Official Study Guide — Eric Ligman, Wiley, Latest Edition.
4.	Practical Cloud Security: A Guide for Secure Design and Deployment — Chris Dotson, O'Reilly Media, Latest Edition.

23CY902	PENETRATION TESTING	1/0/4/3
Nature of Course	F (Theory Programming)	
Prerequisite: -		

Course Objectives:

1	To introduce students to ethical hacking principles, attack methodologies, and threat modeling.
2	To enable learners to identify and exploit operating system, network, and web application vulnerabilities.
3	To train students in using penetration testing tools and techniques to simulate real-world cyberattacks.
4	To empower learners to evaluate and bypass security controls like firewalls, IDS/IPS, and authentication mechanisms.
5	To enable learners to develop GenAI-based scripts for automated penetration testing and real-time vulnerability detection

Course Outcomes:
Upon completion of the course, students shall have ability to

C902.1	Understand core ethical hacking concepts, vulnerabilities, and enumeration techniques	[U[
C902.2	Perform targeted exploitation of operating systems, networks, applications, and web services.	[AP]
C902.3	Evaluate and optimize tools for penetration testing, privilege escalation, and post-exploitation.	[AN]
C902.4	Evaluate and enhance security control effectiveness using standard frameworks and GenAI tools.	[AN]
C902.5	Design and implement a comprehensive penetration testing project simulating real-world threat scenarios.	[AP]

FUNDAMENTALS OF EXPLOITATION **5**

Vulnerability Types and Exploits, System Enumeration Techniques, Operating System Attacks, Malware Threats and Payloads, Privilege Escalation Methods,

WIRELESS EXPLOITS AND APPLICATION EXPLOITATION TECHNIQUES **5**

Network Scanning and Mapping, Exploiting Network Vulnerabilities, Wireless Network Attacks, Bluetooth and Mobile Exploits, Sniffing, Spoofing, and Hijacking. Web Application Vulnerabilities, SQL Injection and XSS, API and Webhook Exploits, Web Shells and Backdoors, Bypassing Security Controls

AUTOMATION AND GAP ANALYSIS **5**

GenAI for Target Enumeration, Automated Exploit Scripting, Honeypots and Detection Evasion, Security Controls Mapping, NIST, CIS Benchmark Evaluation. Project and Summative Assessment – Learners will execute a full-scale penetration test project that includes reconnaissance, exploitation, automation using GenAI, and security control evaluation.

TOTAL PERIODS: 15

Lab Component:

1	Perform enumeration and vulnerability scanning using Nmap and Nikto.
2	Exploit an operating system vulnerability using Metasploit on Kali Linux.
3	Bypass user authentication using password-cracking tools like Hydra or John the Ripper.
4	Exploit a web application vulnerability such as SQL Injection or Cross-Site Scripting (XSS).
5	Simulate privilege escalation in a Linux or Windows environment.
6	Capture and analyze network traffic using Wireshark for session hijacking.
7	Exploit a vulnerable wireless network using tools like Aircrack-ng.
8	Develop a basic GenAI-assisted Python script for automated reconnaissance and scanning.

9	Map enterprise security controls and analyze configuration gaps using OWASP and CIS benchmarks.	
10	Conduct a full-cycle penetration testing simulation including reporting and recommendations.	
	TOTAL (Lab) PERIODS	**60**
	TOTAL PERIODS	**75**

TEXT BOOKS:

1	The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws — Dafydd Stuttard, Marcus Pinto, Wiley, 2nd Edition.
2	Penetration Testing: A Hands-On Introduction to Hacking — Georgia Weidman, No Starch Press, Latest Edition.
3	Kali Linux Revealed: Mastering the Penetration Testing Distribution — Raphael Hertzog, Jim O'Gorman, Mati Aharoni, Offensive Security Press, Latest Edition.

REFERENCE BOOKS:

1	The Hacker Playbook 3: Practical Guide To Penetration Testing — Peter Kim, Secure Planet LLC, Latest Edition.
2	Hacking: The Art of Exploitation — Jon Erickson, No Starch Press, 2nd Edition.
3	Metasploit: The Penetration Tester's Guide — David Kennedy, Jim O'Gorman, Devon Kearns, Mati Aharoni, No Starch Press, Latest Edition.
4	Nmap Network Scanning: The Official Nmap Project Guide to Network Discovery and Security Scanning — Gordon Fyodor Lyon, Insecure.Com LLC, Latest Edition.
5	Advanced Penetration Testing: Hacking the World's Most Secure Networks — Wil Allsopp, Wiley, Latest Edition.
6	OWASP Testing Guide — OWASP Foundation, Latest Version (Open Source).
7	Relevant GenAI in Cybersecurity Reports and Tools — Industry White Papers, Latest.

23CY903	SECURITY OPERATIONS OF INFORMATION SYSTEMS	1/0/4/3
Nature of Course	F (Theory Programming)	
Prerequisite: -		

Course Objectives:

1	To introduce foundational principles of incident response and security operations in modern information systems.
2	To train students to detect, respond to, and recover from various cybersecurity threats using real-time monitoring tools.
3	To equip learners with forensic investigation methods for effective threat identification and evidence preservation.
4	To enable students to apply and assess cybersecurity controls using industry frameworks like NIST and MITRE ATT&CK.
5	To guide learners in generating and automating incident response playbooks using GenAI and other modern tools.

Course Outcomes:

Upon completion of the course, students shall have ability to

C903.1	Describe the core concepts of incident response, policies, and operational security procedures	U
C903.2	Apply containment, eradication, and recovery strategies to manage cybersecurity incidents.	AP
C903.3	Investigate and analyze digital threats using forensic techniques and open-source tools.	AN
C903.4	Evaluate and improve GenAI-driven incident response playbooks based on threat scenarios.	AN
C903.5	Develop and present a functional SOC dashboard or case study applying threat detection and mitigation frameworks.	AP

FOUNDATIONS OF INCIDENT RESPONSE **5**

Incident types and lifecycle, Policy creation and compliance, Roles and responsibilities in IR, Stakeholder communication plan

CONTAINMENT RECOVERY TACTICS AND FORENSICS INVESTIGATION SECURITY CONTROLS **5**

, Initial response procedures, Containment strategies, Eradication methods, Recovery and post-incident activities. Log collection and preservation, Host and network forensics, Chain of custody, Timeline analysis and documentation.

GENAI INTEGRATION **5**

Threat intelligence mapping, SIEM dashboards and alerts, Security control implementation, Playbook generation using GenAI. Develop and present an end-to-end SOC use-case scenario integrating SIEM, forensic investigation, and automated GenAI-based playbook.

TOTAL PERIODS: 15

Lab Component:

1	Configure and deploy Wazuh SIEM and perform basic log ingestion.
2	Create an incident response policy document tailored to a small enterprise.
3	Simulate a containment and eradication scenario using Linux or Windows systems.
4	Collect and preserve forensic evidence from a compromised virtual machine.
5	Use Wazuh to generate alerts for brute-force attacks and unauthorized access.
6	Integrate MITRE ATT&CK mappings into a SIEM dashboard for threat tracking.
7	Develop a basic time-series chart from logs using Wazuh or ELK stack.
8	Generate a GenAI-assisted incident response playbook for ransomware.

9	Conduct a network traffic investigation and produce a report.	
10	Design a custom SOC dashboard displaying security control effectiveness	
	TOTAL (Lab) PERIODS	**60**
	TOTAL PERIODS	**75**

TEXT BOOKS:

1	Incident Response & Computer Forensics — Jason Luttgens, Matthew Pepe, Kevin Mandia; McGraw-Hill Education; 3rd Edition.
2	The Practice of Network Security Monitoring: Understanding Incident Detection and Response — Richard Bejtlich; No Starch Press; Latest Edition.
3	Cybersecurity Incident Response: How to Contain, Eradicate, and Recover from Incidents — Eric C. Thompson; Packt Publishing; Latest Edition.

REFERENCE BOOKS:

1	Blue Team Handbook: Incident Response Edition — Don Murdoch; CreateSpace Independent Publishing; Latest Edition.
2	Digital Forensics and Incident Response: Incident response techniques and procedures to respond to modern cyber threats — Gerard Johansen; Packt Publishing; 3rd Edition.
3	NIST Computer Security Incident Handling Guide (SP 800-61 Rev. 2) — National Institute of Standards and Technology; Free Public Document; Latest Revision.
4	MITRE ATT&CK® for Enterprise: Design and Implementation Guide — MITRE Corporation; Open Source White Papers & Official ATT&CK Documentation.
5	Practical Threat Intelligence and Data-Driven Threat Hunting — Valentina Costa-Gazcon; Packt Publishing; Latest Edition.
6	Wazuh Documentation and ELK Stack Official Guides — Official Open Source Resources; Latest.

SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

An Autonomous Institution | Approved by AICTE | Affiliated to Anna University
Accredited by NAAC with A++ Grade, Kuniamuthur, Coimbatore - 641008

nirf 83rd Rank

SUSTAINABLE DEVELOPMENT GOALS

SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

An Autonomous Institution | Approved by AICTE | Affiliated to Anna University

Accredited by NAAC with A++ Grade, Kuniamuthur, Coimbatore - 641008

E: info@skcet.ac.in | W: www.skcet.ac.in